

An Empirical Analysis of Financial Markets: An Econophysics Approach

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3 Abstract

4 We develop a physics-inspired microstructure model for digital asset mar-
5 kets that processes Level-3 limit-order-book (LOB) events in real time. Treat-
6 ing orders as particles and order activity as system dynamics, we derive in-
7 terpretable measures, kinetic energy and momentum, in order to forecast
8 short-horizon volatility and direction. To scale beyond top-of-book snap-
9 shots, we introduce *Active Depth*, a data-driven procedure that selects the
10 most informative LOB layers from event-time flows. Across cryptocurrency
11 datasets, the proposed measures outperform VPIN and Order Flow Imbal-
12 ance in both volatility and directional forecasting, and serve as high-value
13 features that improve DeepLOB, LSTM, and Transformer baselines. The
14 approach is training-free, latency-aware, and deployable at depth, offering
15 transparent signals for trading, market surveillance, and risk management.
16 We discuss implications for digital asset and provide studies validating *Active*
17 *Depth* and the incremental value of physics-informed features. Our results
18 highlight how econophysics can complement machine learning to deliver in-
19 terpretable, computationally efficient prediction in financial markets.

20 *Keywords:* Econophysics, Market Microstructure, Limit Order Book,
21 Machine Learning, Cryptocurrency

1. Introduction

We study digital asset markets through a physics-inspired microstructure model that ingests Level-3 limit-order-book (LOB) events in real time. Our objective is to forecast short-horizon volatility and direction while providing interpretable signals that can be operationalized in trading, market-surveillance, and risk-management systems. The paper makes three contributions.

Implication on asset pricing for digital asset markets. First, this work sits squarely within Digital Finance. It addresses data-intensive markets (crypto), develops AI-compatible features (physics-informed signals that lift deep models), and targets operational needs in risk management, market surveillance, and digital governance (transparent real-time indicators, depth-aware scaling). By linking econophysics with deployable analytics on Level-3 data, the paper speaks to the special issue's focus on big data, ML/AI, digital currencies/blockchain, and the economic-regulatory challenges of digital markets.

We establish novel benchmarks for the predictive accuracy of cryptocurrency asset price volatility and expected return utilising Level 3 order book data. The accuracy of our model is justified by high predictive R-squared values and prediction accuracy compared to previous literature, illustrating its robustness across different market conditions. Our model bypasses the need for training samples or parameter fitting since it takes full advantage of the market microstructure information inherently enclosed in the Level 3 order book data. Challet and Stinchcombe (2003), Eisler et al. (2012) and Cont et al. (2014) studied the price impact of market order submissions, and limit order submissions and cancellations that contained in the Level 3 order book data, and discussed how these actions affect the limit order book (LOB). This information is adequate for estimating the impact of both current and forthcoming orders on the order book dynamics and, subsequently, price dynamics. We underline significant economic implications for investors leveraging these predictive measures, which display a considerable edge over conventional techniques such as the Volume-Synchronized Probability of Informed Trading (VPIN) and the Order Flow Imbalance (OFI) when predicting volatility and expected asset returns.

Volatility and return prediction carry tangible economic significance. The principal objective of asset pricing revolves around comprehending the patterns of expected return and its accompanying volatility, as presented by

French et al. (1987). Even if expected returns were observed completely, the requirement for theories explaining their behaviour still persists and the same for the empirical analysis validating them. However, the tasks of measuring volatility and return are well-known intricate: at a lower frequency time scale (like hourly or daily), return variations are primarily influenced by unpredictable news events that impact prices. Upon examination of identical datasets, distinct volatility estimators can manifest disparate characteristics. Notably, according to Cont et al. (2014), these estimators exhibit varied intra-day patterns in different underlying assets. On the other hand, at high-frequency trading scales, they are shadowed by the market microstructure noise generated by market participants, including market makers, informed traders, and noise traders.

Cont et al. (2014) proved that the order book depth deeper than best bid and best ask contain information relevant to the price dynamics. Based on this empirical finding, a recent strand of literature attempted machine learning methods to capture the volatility and expected return of assets. Doering et al. (2017) developed convolutional neural networks (CNN) to predict price and its volatility. Kercheval and Zhang (2015) employed Support Vector Machine (SVM) on the order book data. Sirignano (2019) and Sirignano and Cont (2019) developed a deep Long Short-Term Memory (LSTM) model to study the snapshot data of the order book. However, due to the restriction of computing power and model complexity of machine learning models, they are not feasible to study the information in deeper depths of the order book. Our proposed physics-based model defines the concept of Active Depth to distinguish the order book depths relevant with a simplicity in model complexity. The model interpretability is another shortcoming of the machine learning methods labeled as ‘black boxes’ because of the difficulty to understand the precise reasoning behind them, while the physics-based model provides a straightforward intuition to explain the order book dynamics by modelling it as a physical system.

Market microstructure noise is another important topic to consider as it captures various of frictions intrinsic to the trading process. As argued by Ait-Sahalia and Yu (2008), these noises include the bid-ask bounces, discreteness of price fluctuations, differences in trade sizes, the informational content of price changes, strategies of the order flow, and inventory management effects. Eliminating the impact of these noises during investigating the asset price is meaningful for the asset pricing practices, particularly in the high-frequency trading markets.

We provide a novel technique to model the order book dynamics that is possible to bypass hence eliminate the impact of both market news and market microstructure noises as they arise from the time series of asset price, volume, bid-ask spread or any other explanatory variables generated by order book dynamics. As a result, our research acts as a foundation upon which future studies can be built, moving the field towards a more comprehensive understanding of the mechanisms that dominate asset pricing and, by extension, the broader financial market dynamics.

Furthermore, we conduct an extensive benchmarking exercise comparing our Physical Model to state-of-the-art deep learning architectures, including Deep LSTM, DeepLOB, and a custom LOB-Transformer. The results show that our model achieves competitive accuracy with no need for training, while also significantly enhancing the performance of these models when used in conjunction. This highlights the robustness and utility of physics-inspired measures in high-frequency market prediction tasks.

A physics-informed microstructure model for digital asset markets. Second, we integrate microstructure-based asset pricing with a physics-informed representation of LOB event flows, yielding interpretable, deployment-ready signals for digital markets. While asset pricing traditionally focuses on determining the intrinsic value of assets, an understanding of the market microstructure – the system through which the trading of assets takes place – is crucial to comprehend how the supply and demand for assets are matched and how prices are ultimately discovered.

Bouchaud et al. (2002) along with Potters and Bouchaud (2003) elucidated the statistical intricacies inherent to the order book. These empirical investigations served as the foundation of 'zero-intelligence' models. Such models posited that the 'stylised facts' are an emergent outcome of order flow properties and the intrinsic structure of the order book, without the need for the 'rationality' assumptions. In their studies on limit order book, Bouchaud et al. (2002) and Potters and Bouchaud (2003) offered a systematic exploration of empirical properties, and developed models based on econophysics that linked the financial markets with physical systems.

The increasing interest in applying physical models to finance, both in academia and in the industry, has yielded empirical studies in the field of asset pricing and the field of Econophysics. Bouchaud et al. (2003) examined the complex interplay between order flow and price fluctuations, drawing analogies with the external stimuli in physical systems. Their insights unveiled the market 'randomness'. Farmer and Lillo (2004) through rigorous analy-

sis, highlighted a connection between power-law tails in price changes and order flow dynamics, aligning with phenomena observed in physics. Farmer et al. (2006) explored the effects of supply-demand on price movements, contrasting with the Efficient Market Hypothesis. Their findings drew parallels with long-range correlations in physical systems. Borland (2012) looked into the statistical pattern during market panics, by harnessing the concept of self-organizing systems from physics. Inspired from these studies in Econophysics, our analysis aim to provide a detailed investigation of the utility of a physical model applied to the challenging problems of modelling the LOB.

Bak et al. (1997) offered a distinctive perspective on price dynamics within a multifaceted agent stock market. Drawing parallels with physical systems, orders were conceptualised as particles on a price trajectory, where intersections resulted in transactions. Yura et al. (2014) conceptualised orders as fluid particles, using an interaction range defined by order activity to segment the order book into layers. Yura et al. (2015) expanded this model, linking price movement velocity to changes in the inner layer's time-averaged limit orders, and introduced a financial Knudsen number.

Approaching the complex structure of the limit order book from a econophysics perspective allows us to view the orders as interactive elements, and the price fluctuation of an asset as a macroscopic property of this complex system. The historical price data merely encapsulates past alterations in this macroscopic property, analogous to the temperature in a fluid system. Nonetheless, examining the evolution of a fluid system necessitates knowledge of inherent physical parameters such as density, mass, and velocity. These parameters cannot be seized from historical temperature data alone; they require an understanding of the complex system. The key of the argument lies in the fact that the limit order book system is vastly more information-rich compared to historical price data.

The current literature of asset pricing does not address to this point but merely studies the time series of asset price, while econophysics provides the framework and techniques to build such models. By treating the limit order book as a physical system, we can tap into a deeper, more macroscopic level of information. This approach, akin to the methods employed in econophysics, allows us to better understand the underlying mechanisms that drive the dynamics of asset prices. By recognizing that price movements are a macroscopic feature of a more complex system of interacting orders, we can gain new insights into the asset pricing process. This perspective opens up novel avenues for research and presents opportunities for the development

173 of more robust models for understanding the price discovery and predicting
174 asset prices.

175 We employ the term ‘physical model’ to refer to a novel method that
176 mimics the dynamics of the order activities. We introduce a econophysics
177 perspective to model the limit order book, drawing an analogy between fi-
178 nancial markets and physics. It constructs a microscopic model of the orders,
179 conceptualising them as particles within the complex system of the limit or-
180 der book. Order submissions are interpreted as particles entering the system,
181 order cancellations as particles exiting the system, and each order transac-
182 tion as a particle annihilation. This allows us to calculate the change in
183 ‘kinetic energy’ and ‘momentum’ within the system and estimate the im-
184 pact of each activity on the LOB. Our method considers the different quoted
185 prices of each single order upon submission and cancellation, estimates the
186 impact of each order activity on the order book, and utilises the microscopic
187 dynamics of order book activities to construct a systemic description of mar-
188 ket behaviour. We evaluate their efficacy in predicting price changes in the
189 Bitcoin and LUNA markets. Our findings suggest that these new measures
190 offer enhanced predictive power and capacity to accurately assess the market
191 behaviour, price volatility and return of the cryptocurrencies, outperforming
192 existing models such as the VPIN and OFI measure. This approach also
193 enables us to visually highlight events occurring within the high-frequency
194 trading market. Based on the current literature, few analytical tools har-
195 ness the vast amount of data in this domain to effectively illustrate financial
196 market phenomena.

197 In our analysis, we find that the predictive power of these measures, when
198 analyzed through the lens of our physical model, surpasses that of the bench-
199 mark techniques. Our findings show an improvement in the ability to forecast
200 price volatility and return in the asset. Moreover, the physical model’s suc-
201 cess in addressing the inefficiencies of traditional prediction models leads us
202 to our next empirical fact: the use of physical models to analyse market mi-
203 crostructure is viable when complemented with accurate description of the
204 order book complex system measured by the physical models. This empirical
205 research serves to justify the role of such models in the broader architecture
206 of the financial markets, which offers a significant advantage in capturing the
207 complexity of the order book dynamics that the traditional methods often
208 fail to accomplish.

209 However, it’s important to note that while our physical model signifi-
210 cantly enhances our understanding of asset returns, it is fundamentally a

211 measurement tool. It merely explains the process that the limit and market
212 orders discover the price and impact its dynamics, while it does not provide
213 insights into the economic mechanisms that drive these activities. As such,
214 while the physical model can significantly describe market behaviour and
215 enhance prediction, it must be complemented with economic theories that
216 investigate the underlying mechanisms driving asset returns. This remains an
217 exciting direction for future research, as there is an emerging literature that
218 combines these advanced analytical tools with economic theories to provide
219 a more comprehensive understanding of asset returns.

220 **Active Depth: data-driven depth selection for scalable real-time**
221 **systems.** While top-of-book dynamics drive immediate price formation,
222 deeper levels encode broader market sentiment, hidden liquidity, and strate-
223 gic placement. Active Depth endogenously selects the LOB region where the
224 reacted order volume is able to predict price and volatility, enabling depth-
225 aware modeling with efficient computation

226 Bouchaud et al. (2002) emphasised the importance of understanding the
227 full depth of order books and not just the top layers. Their models, which
228 built on econophysics, revealed the intricate patterns and correlations across
229 different layers, underscoring the significance of considering deeper levels
230 when analysing price dynamics. Zovko and Farmer (2001) highlighted the
231 nuanced ‘patience’ traders demonstrate, suggesting that order placement is
232 not just concentrated near the best bid and ask, instead, many traders placed
233 orders at deeper levels and the distribution decayed as a power law. Through
234 simulation analysis, Smith et al. (2003) provided a statistical framework to
235 describe the continuous double auction mechanism. Their theory encapsu-
236 lated the significance of interactions at varying depths in the order book. By
237 illustrating how orders at different layers influence price determination, they
238 accentuated the importance of understanding the full spectrum of the order
239 book for accurate price prediction. Chiarella and Iori (2002) brought forth
240 insights into the microstructure dynamics of double auction markets. Their
241 agent-based model, with traders influenced by factors like chartists, funda-
242 mentalists, and noise, underscored that the depth of the order book was not
243 static but evolved based on these influences. They advocate for an in-depth
244 examination of the order book, suggesting that the deeper levels can provide
245 substantial insights into forthcoming market movements.

246 However, due to the restriction of computing power and model complex-
247 ity, it is impossible to study the information in all the depths of the order
248 book. Market makers, informed traders, noise traders and other market par-

participants participate in the market by submitting and cancelling the orders. The order book dynamics is impacted by these order activities. As a result, it is possible to find the depths that are encountering such activities. Based on this inspiration, we proposed a method to define and calculate the concept of active depth, which distinguish the depths that are active and contains the relevant information to impact the order book.

In conclusion, the physical model, with its purely empirical approach based on limit order book data, provides a fresh perspective that transcends the traditional assumptions of empirical and behavioral finance. This methodology offers a more nuanced and comprehensive analysis of market dynamics, heralding a new direction in financial market research. However, it is important to acknowledge that while this approach provides valuable insights, it does not entirely replace the need for understanding the underlying economic mechanisms or equilibrium. Complementary use of empirical, behavioral, and the physical model may provide a more holistic perspective of financial markets and asset pricing.

Beyond theoretical elegance, the model's practicality is validated through systematic experiments showing strong generalisation across prediction horizons, minimal latency, and high interpretability. These qualities make the Physical Model suitable not only as a standalone predictor but also as a complementary module in modern machine learning pipelines for trading systems.

The structure of this article is as follows. The review on the current related literature is given in Section 2. In Section 3, we overview the source and format of the cryptocurrency order book data we used for our experiments. A physical model to describe the limit order book and the order activity is presented in Section 4, as well as the definition and calculation of the prerequisite concept called active depth. In Section 6 the kinetic energy measure from the physical model is compared to the VPIN method on predicting the volatility. In Section 7 another measure of momentum is discussed in a comparative analysis to the Order Flow Imbalance method. In Section 8, we benchmark the Physical Model against deep learning models and show how it improves predictive performance when integrated with them.

2. Literature Review

We here give an overview of existing financial models and methodologies, focusing on those relevant to model the financial market with physics and

those predicting price volatility and impact.

Zero-intelligence econophysics models. A key theoretical foundation for our work comes from zero-intelligence models, which assume that market participants submit orders randomly rather than strictly optimizing utility functions. These models have proven surprisingly effective in capturing a range of empirically observed market phenomena. Notably, Smith et al. (2003) developed a statistical theory of the continuous double auction, demonstrating that many stylised facts about prices and order flows emerge even under zero-intelligence conditions. Farmer et al. (2005) further showed that zero-intelligence models can exhibit nontrivial predictive power, highlighting that certain regularities of limit order markets do not necessarily require strong assumptions about rationality or strategic behavior. They refined the modelling of order arrival, price movement, and implied supply-demand relationships in a continuous double auction context. They established functional relationships between order-arrival rates and market statistical characteristics, as well as price movements and order flow. Upon testing this model with trading data from different stocks, they found a shared price-impact curve, suggesting a universal statistical law. Lillo and Doyne Farmer (2005) demonstrated that the multidimensional stochastic process of bid-ask spread exhibits specific statistical attributes, and that fluctuations in the state of the limit order book drive price movements.

Lillo et al. (2003) conducted data analysis on the determinants of the price impact of individual transactions, accounting for factors such as volume and underlying stock characteristics. To standardise across different liquidity levels, they re-scaled the mean price change and volume, resulting in a uniform price-impact curve across various stocks. Their findings suggested a universal statistical law governing fluctuations from supply-and-demand equilibrium across different financial assets. They postulated the efficacy of modelling human activity as stochastic in certain contexts over rational models. Building on these insights, Daniels et al. (2003) modeled the price formation process, assuming random order placement and cancellation, which allowed them to predict price movements successfully.

The aforementioned studies provide evidence that the state of the limit order book and movement of limit orders follow stochastic processes, revealing certain statistical laws governing the limit order book. Consequently, it becomes viable to study the rules governing order activities using econophysics. Beyond zero-intelligence approaches, our paper also draws on the larger econophysics literature, which connects statistical physics methods to

financial market dynamics. Bouchaud et al. (2009) provided a foundational survey on how markets absorb order flow imbalances over time, emphasizing that imbalances in supply and demand can persist and exhibit long-range correlations. Their work underscores the idea that market price dynamics often arise from local interactions among heterogeneous agents, an insight that aligns with our analogy of treating orders as interacting ‘particles’ within an order book.

Yura et al. (2014) studied the LOB by drawing a parallel between financial market dynamics and Brownian motion. They segmented the order book into layers based on an ‘interaction range’ defined by order activity. Consequently, they modeled the bid-ask spread and the nearest depths with the highest temporary volume as a colloidal financial Brownian particle. Their work provided valuable insights on investigating how order book structures drove price dynamics with econophysics. Li et al. (2023) modelled the orders as particles and calculated the LOB system’s momentum in the defined passive area, which they use to recognise the anomalous patterns and detect market manipulation activities. They showcased the potential of physical models on Level 3 order book data which may not be discernible using conventional financial econometrics tools. Although this work did not study the volatility and expected return, Li et al. (2023) offered an innovative way of viewing the LOB and analysing the behaviours of market participants.

Order flow and microstructure metrics. Easley et al. (2011) provided an in-depth analysis of the events of May 6, 2010, known as the ‘flash crash’. The authors argued that the crash was precipitated by automated execution of a large sell order in the E-mini contract, leading to two liquidity crises. They presented evidence that the liquidity problem was slowly developing in the hours and days before the collapse, with high and unbalanced volume but low liquidity. The authors also discussed the role of high-frequency trading (HFT) firms in this event, noting that these firms, which account for over 73% of all U.S. equity trading volume, often act as market makers. The paper introduced a measure of order flow toxicity, VPIN, and demonstrated its predictive power for absolute returns and its potential use as a warning measure for the risk of a liquidity crash. The authors concluded that in markets dominated by high-frequency liquidity providers, unusually high levels of flow toxicity could lead to liquidity providers turning into liquidity consumers, potentially destabilising the market.

In the paper by Andersen and Bondarenko (2014), the authors further critically examined the Volume-Synchronised Probability of Informed Trad-

ing (VPIN) and its variants. They argued against the significance of any specific metric unless it retained significance in formal tests that incorporate readily observable real-time market activity measures, such as trading intensity and implied volatility measures. The authors also discussed the volume clock, the Order Imbalance (OI) measure, and the Trade Classification (TRVPIN) metric. They highlighted the complex interplay of parameters such as the volume bucket, the time bar, the trade classification indicator, and the length of the moving average in determining both the level and the dynamic behaviour of the metric. The authors concluded that these parameters were necessary to gauge the incremental information content of any new metric.

Easley et al. (2021) further demonstrated the application of machine learning algorithms to predict and explain modern market microstructure phenomena. They investigated the efficacy of various microstructure measures, showing that they continued to provide insights into price dynamics in current complex markets. Interestingly, some microstructure features with high explanatory power exhibited low predictive power, and vice versa. The authors also found that some microstructure-based measures were useful for out-of-sample prediction of various market statistics, leading to questions about market efficiency. Their results were derived using 87 of the most liquid futures contracts across all asset classes. The authors suggested that as markets become faster and data more copious, the market microstructure played a critical role in predicting market behavior. They argued that despite the increased complexity of trading strategies and the empirical measures of microstructure, machine learning techniques can play an important role in the evolution of microstructure research.

Regarding to the market microstructure analysis based on the order flow, Cont et al. (2014) empirically investigated the influence of order book events on equity prices using NYSE TAQ data. It proposed a simplified model using a single variable, the Order Flow Imbalance (OFI), to linearly explain mid-price changes, demonstrating robustness across various stocks and timescales. Based on this work, Shen (2015) presented a compelling exploration of the Volume Order Imbalance (VOI), which provided a robust framework for understanding market dynamics and offers valuable insights into the supply and demand analysis on market microstructure level. Cartea et al. (2018) provided an in-depth exploration of how trading strategies can be improved by incorporating signals from the order book. Their research offered valuable insights into the application of high-frequency order book data in strategy optimization, thereby broadening the understanding of market microstruc-

ture and its implications for trading strategies. Their rigorous mathematical approach and insightful findings underscored the importance of incorporating granular market data to enhance trading efficacy.

Machine learning methods for financial markets. In recent years, applying machine learning approaches on finance is an emerging topic while there are relatively few that using such approaches to study the LOB. Nevmyvaka et al. (2006) introduced machine learning to the LOB by applying reinforcement learning on optimal order execution. Kercheval and Zhang (2015) applied support vector machines (SVM) to predict the direction move of the mid-price for stocks. Sirignano (2019) argued that neural networks outperforms SVM as well as other machine learning methods for LOB as they are more capable of dealing with high-dimensional data and capturing nonlinear relationships, while SVM can only provide binary classification instead of indicate probabilities. Sirignano and Cont (2019) further developed a deep neural network (DNN) model with Long Short-Term Memory (LSTM) units. They trained an universal model with the LOB data of about 500 NASDAQ stocks to predict the direction move of price. Yin and Wong (2023) engineered deep convolutional neural network model to seize the information from the LOB data at tick frequency of 0.5 second. After exploiting such information, the performance of their trading algorithm proved that the LOB data contained rich information for liquidity, price prediction and trading signal generation. Shavandi and Khedmati (2022) and Huang et al. (2024) developed deep reinforcement learning model and proved them efficient on optimizing algorithmic trading strategies.

While our research emphasises an interpretable physics-based approach, the growing literature on deep learning methods for limit order books offers a different angle on capturing microstructure dynamics. For instance, Tsantekidis et al. (2017) investigated how deep learning architectures, including recurrent and convolutional networks, can detect price-change signals or volatility patterns embedded in high-frequency LOB data. Similarly, Zhang et al. (2019) explored deep convolutional neural networks trained on order book snapshots (known as DeepLOB). Their framework automatically learns predictive features from multi-level bid and ask data, achieving strong forecasting performance on short-term price moves. Both works illustrate that well-designed neural networks can uncover complex, nonlinear relationships in trading data—albeit often at the cost of substantial computational resources and reduced interpretability.

In contrast, our physics-based method does not require large-scale GPU

437 training or extensive hyperparameter tuning. By focusing on kinetic energy
438 and momentum constructs, adapted from zero-intelligence and econophysics
439 principles, we obtain interpretable metrics that directly connect to the under-
440 lying order flow. This approach provides a practical alternative or comple-
441 ment to deep neural networks, particularly in high-frequency contexts where
442 computational overhead and model explainability are crucial considerations.
443 Moreover, the model is straightforward and does not rely on extensive pa-
444 rameter fitting or large-scale training. This interpretability is a practical
445 advantage compared to typical ‘black box’ machine learning models.

446 3. Data

447 This section details the data sets used in the study. It describes the source
448 of the data, which includes transaction data for Bitcoin and LUNA, and the
449 time frame covered. It also outlines the steps taken to clean and prepare the
450 data for analysis.

451 Financial markets are typically organised around a central record, the
452 limit order book (LOB), which logs all orders placed by traders. Our study
453 aims to process Level 3 order book data and summarise the complete infor-
454 mation about order activity and order book dynamics. Rather than relying
455 on time-based snapshot data of the order book, we propose a more compre-
456 hensive approach that processes every detail and message of the order book
457 data, thereby preserving all temporal information.

458 These data are considered to be high-frequency trading data (in microsec-
459 onds) sourced from the Coinbase exchange, accessed via Websocket feeds.
460 Due to the nature of the comprehensive physical modelling, this study re-
461 quires large-scale event data to model them with physical particle dynamics.
462 As a result, we study the BTC/USD pair as it has the largest trading volume
463 among existing cryptocurrencies. We also investigated the LUNA/USD pair
464 about its well-known flash crash in May 2022.

465 There are only limited exchanges that provide the Level-3 order book
466 event data, and Coinbase is the most popular exchange hence offers the
467 largest amount of available data demanded by this study. Our Level-3 order
468 book data were procured from the Coinbase exchange via a websocket feed,
469 encompassing Level-3 trading data for the BTC/USD and LUNA/USD cryp-
470 tocurrency pairs. As the most granular form of the order book data, Level-3
471 data includes records of order submissions, cancellations, and matches. In

472 contrast, Level-1 data comprises the Open/High/Low/Close price and vol-
473 ume data for a given period, while Level-2 data incorporates additional in-
474 formation regarding bid and ask order depths.

475 Orders received at a price equal to or worse than an existing order (typ-
476 ically, the best bid/ask) are executed immediately, generating a ‘match’
477 record. Unexecuted or partially executed orders become limit orders in the
478 LOB, creating an ‘open’ record. These limit orders, if not excuted, will re-
479 main on the LOB until they are cancelled, leaving a ‘cancelled’ record. In
480 our dataset, approximately 98.6% of open orders are eventually cancelled.

481 The minimum price precision is 0.01 USD for both BTC/USD and LUNA/USD
482 pairs, and the minimum trading volume unit (order size) is 0.00001 BTC for
483 BTC/USD pair and 0.001 LUNA for LUNA/USD pair. The minimum pre-
484 cision of the recorded timestamp is 0.000001 second (one microsecond). A
485 timestamp is counted when an order activity (or, event) occurs, hence the
486 timestamps of the event-driven records are discrete and not consecutive. For
487 both BTC/USD and LUNA/USD datasets, we grouped the event records
488 by $\Delta t = 0.1$ second basis and aggregated all the events occurred in each
489 $\Delta t = 0.1$ time period. We also recorded the best bid and best ask prices
490 when the last event of each $\Delta t = 0.1$ time period occurs as the best bid/ask
491 prices of the consecutive $\Delta t = 0.1$ time period.

492 The granularity of Level-3 order book data presents a rich tapestry of
493 information, demanding opportune computational strategies. We opted to
494 aggregate events into 0.1-second intervals, representing a careful balance be-
495 tween retaining market activity resolution and optimising computational ef-
496 ficiency. The empirical assessments confirmed that this aggregation granu-
497 larity preserved the integrity of significant market patterns and trends.

498 Section 5 analysed cryptocurrency market behaviour by investigating
499 Level-3 trading data for the BTC/USD pair between 20/04/2022 23:00-00:00
500 and the LUNA/USD pair between 11/05/2022 16:00-20:00, when the LUNA
501 crash occurred. In the LUNA/USD dataset from 11/05/2022 16:00-17:00, a
502 total of 248,796 order submission and cancellation records were identified,
503 with 233,134 occurring within the active area, 8,302 within the passive area,
504 and 7,360 outside. Consequently, about 97% of order submissions and can-
505 cellations transpired within the active area, and 3% within the passive area.

506 In Section 4.2, 6 and 7, we investigated the Level-3 order book data for the
507 BTC/USD pair during 02:00-03:00 20/04/2022, 15:00-16:00 on 28/11/2022,
508 and for the LUNA/USD pair during 16:00-20:00 on 11/05/2022, 02:00-03:00
509 on 12/05/2022, while the LUNA was facing the significant fall. The original

510 datasets are composed of two datasets - full channel data and ticker channel
511 data. The full channel encompasses all of the Level-3 order book data dis-
512 cussed previously, and is event-driven, with every order activity (submission,
513 cancellation, and matches) leaving a record. On the other hand, the ticker
514 channel contains snapshot information of the order book, including the best
515 bid price and best ask price, with a new record in this channel created each
516 time a match occurs. However, the aggregated quoted volume of the best
517 bid and best ask price was available for the BTC/USD data, but not for the
518 LUNA/USD data. There were more than 2,700,000 lines of records in the full
519 channel data and 47,000 in the ticker channel for each one-hour BTC/USD
520 dataset, while the numbers were about 480,000 and 35,000 for LUNA/USD.

521 We also used both the Level-3 order book data and LOB snapshot data
522 for the BTC/USD pair during the 12-hour span between 16/09/2023 02:00 -
523 16/09/2023 14:00 to train the Deep LSTM model, the DeepLOB model, and
524 an LOB-Transformer model to analyse the capability of the physics-based
525 model in Section 8. This data set contained 9,839,922 samples of Level-3
526 order book data and 2,018,900 samples of LOB snapshots.

527 The high-frequency nature of the dataset resulted in a significantly large
528 amount of data points, even for a relatively short timeframe. The usage,
529 time span, and nature of the data we have used in this study are listed in
530 Table 1.

Table 1: Summary of datasets used throughout the study

Dataset	Section and Usage	Time Span	Event Data	Snapshot Data
BTC/USD (I)	Section 5 - empirical market behaviour	20/04/2022 23:00-00:00	✓	✗
LUNA/USD (I)	Section 5 - empirical market behaviour	11/05/2022 16:00-20:00	✓	✗
BTC/USD (II)	Section 4.2 - calculation on <i>Active Depth</i>	28/11/2022 15:00-16:00	✓	✗
	Section 6 - comparison with VPIN on volatility			
	Section 7 - comparison with OFI on price prediction			
LUNA/USD (II)	Section 4.2 - calculation on <i>Active Depth</i>	12/05/2022 02:00-03:00	✓	✗
	Section 6 - comparison with VPIN on volatility			
	Section 7 - comparison with OFI on price prediction			
BTC/USD (III)	Section 8 - Benchmarking with deep models	16/09/2023 02:00-12:00	✓	✓

531 Why Level-3 data?

532 Throughout the empirical chapters we rely on Level-3 (order-event) feeds
533 rather than the coarser Level-1 (best quote) or Level-2 (depth snapshot)
534 alternatives. Two considerations motivate this choice.

535 Requirement for event granularity: The Physical Model treats every order
536 as a particle whose displacement, velocity and momentum must be observed

at the moment of submission or cancellation. These quantities are not recoverable from aggregated depth messages; once orders are bucketed by price level, the exact path that each quote has travelled through the book is irretrievably lost.

Ability to diagnose manipulation at source: Rule-based surveillance works reasonably well when abuse manifests only in executed trades, but it struggles whenever abusive strategies exploit the order-submission process itself (for example, spoofers layering quotes far from the touch but cancelling milliseconds before execution). Level-3 order book data feeds are the only public record that preserves those additions and withdrawals before they influence the best bid or ask. Without this detail a spoofing episode may be indistinguishable from innocuous quote flickering driven by market making algorithms.

Level-3 order book data feeds therefore supply the minimum information set required to compute physical variables reliably and to attribute any detected anomaly to an identifiable sequence of messages. Lower-resolution data may still be useful for descriptive statistics, but they cannot support the particle-based mechanics that underpins this work.

4. Methodology

The Physical Model, with its purely empirical approach based on limit order book data, provides a fresh perspective that transcends the traditional assumptions of empirical and behavioural finance. This methodology offers a more nuanced and comprehensive analysis of market dynamics, heralding a new direction in financial market research.

4.1. The Physical Model

In this section, we will propose a model for the LOB inspired by concepts from physics, which models the LOB as a physical system where the conservation of momentum and energy can be applied. We will extend these concepts to introduce and define the novel measures such as the Active Depth, total volume of reacted order, and kinetic energy of the LOB.

In the LOB, the best bid and best ask prices are represented by b_M and a_M , and the market midprice is subsequently denoted by $z_{mid} = \frac{b_M + a_M}{2}$. Additionally, the market match price p^M is noted whenever a trade execution occurs. For a given time t , the market midprice and match price are denoted by $z_{mid}(t)$ and $p^M(t)$ respectively. The quantity of limit orders at a specific

572 quoted price x is denoted by $N(x, t)$ which is invariably non-negative. Over
 573 the interval $[t - \Delta t, t]$, the *velocity* of the match price, $v^M(t)$, is calculated
 574 by the rate of change in the match price with $v^M(t) = \frac{p^M(t) - p^M(t - \Delta t)}{\Delta t}$.

575 It was observed that order activities predominantly cluster around the
 576 market midprice. A so-called *active area* is identified as an extended range
 577 at a particular depth within the LOB, oriented around the bid-ask spread as
 578 illustrated in Figure 1. The dynamics of the active area closely mirror those
 579 of the market midprice; within this range, order submissions and cancella-
 580 tions exhibit elevated activity, while beyond this range, limit orders generally
 581 remain stationary. Consequently, the *active bid price* and *active ask price*
 582 are derived as the best bid and best ask prices adjusted by the Active Depth
 583 defined in Section 4.2.

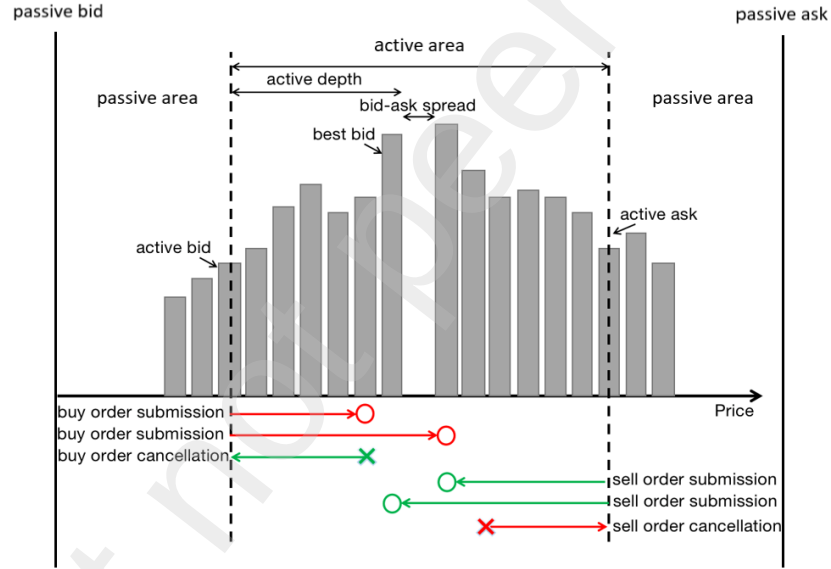


Figure 1: Active and passive areas of the order book

584 In this study, orders are conceptualised as physical particles moving on
 585 a one-dimensional axis. The order size corresponds to the particle mass,
 586 while its displacement is akin to the distance of particle movement. Over a
 587 sampling period Δt , the velocity $v(t)$ of an order is computed as the quotient
 588 of its displacement Δp and the elapsed time Δt . The order velocity is only
 589 defined within the active area, bounded by $[b_M - \alpha, a_M + \alpha]$ where α is the
 590 Active Depth which will be defined in Section 4.2.

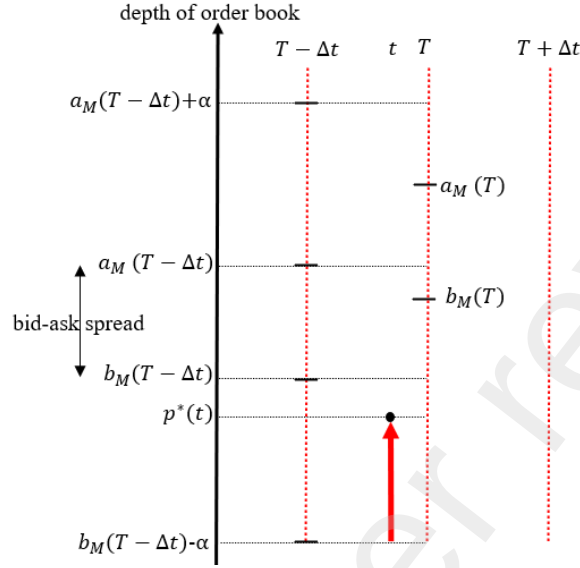


Figure 2: Motion of a limit buy order

During each sampling interval $(T - \Delta t, T]$, we define the velocity of a limit buy order as the rate of change in its quoted price relative to the active bid price $b_M - \alpha$. Specifically, the velocity of a limit buy order quoted at price $p^*(t)$ at time t reflects its movement from price depth $b_M - \alpha$ to $p^*(t)$ within this period, encapsulating the intensity and direction of order activity in the LOB. In other words, for a specific limit buy order of size s submitted to its quoted price $p^*(t)$ at time t during the sampling period $(T - \Delta t, T]$, $t \in (T - \Delta t, T]$, the order's velocity can be computed with

$$v(t) = \frac{p^*(t) - (b_M(T - \Delta t) - \alpha)}{\Delta t}. \quad (1)$$

It is noteworthy that for limit buy orders exceeding the best ask price a_M , the effective quoted price becomes $p^*(t) = a_M$ at time $T - \Delta t$.

In contrast, a canceled buy order is considered to move from its initial quoted price $p^*(t)$ to the active bid price, $b_M - \alpha$ at time $T - \Delta t$. Its velocity is thus calculated as:

$$v(t) = \frac{(b_M(T - \Delta t) - \alpha) - p^*(t)}{\Delta t}. \quad (2)$$

604 Similarly, for a newly submitted limit sell order, the velocity is

$$v(t) = \frac{p^*(t) - (a_M(T - \Delta t) + \alpha)}{\Delta t}. \quad (3)$$

605 Here, if the quoted price is below the best bid price b_M , the effective quoted
606 price becomes $p^*(t) = b_M$ at time $T - \Delta t$. For canceled sell orders, the
607 velocity is computed as:

$$v(t) = \frac{(a_M(T - \Delta t) + \alpha) - p^*(t)}{\Delta t}. \quad (4)$$

608 The depth of the order book will be raised to specify the location of limit
609 orders with regard to the inside of the order book (best bid price for buy
610 orders and best ask price for sell orders) at the same side. At time t , the
611 depth of a specific limit order quoted at price p^* is denoted by $\gamma(p^*, t)$ and
612 defined by

$$\gamma(p^*, t) = [b_M(T - \Delta t) - p^*] \quad (5)$$

613 for the bid side, where the buy orders are normally quoted at price lower
614 than the best bid price. And

$$\gamma(p^*, t) = [p^* - a_M(T - \Delta t)] \quad (6)$$

615 for the ask side, where the sell orders are normally quoted at price higher
616 than the best ask price. Note that the depth for a limit order could be
617 negative value in the case that it is quoted between the bid-ask spread.

618 Let $N_\gamma(t)$ denote the total number of limit orders submitted at depth γ
619 of the LOB during time t , and $N_\gamma^c(t)$ for the total number of limit orders
620 cancelled at depth γ at time t . We define the Reacted Orders Volume (either
621 cancelled or submitted) inside depth γ during $T - \Delta t$ to T as

$$\Delta V(\gamma)_{(T-\Delta t, T]} = \sum_{t \in (T-\Delta t, T]} \sum_{\gamma^* = b_M - \gamma}^{a_M + \gamma} \left(\sum_{i=1}^{N_{\gamma^*}(t)} s_i + \sum_{j=1}^{N_{\gamma^*}^c(t)} s_j^c \right) \quad (7)$$

622 where s_i is the order size of the i^{th} submitted limit order quoted at this
623 depth, while s_j^c is the order size of the j^{th} cancelled limit order quoted at
624 this depth. Note that this equation takes discrete values, over the discretised
625 event-based order book data timeframe $t \in (T - \Delta t, T]$, on the order book
626 depth γ^* ranging from the depth $b_M - \gamma$ at best bid price minus the given

value γ , to the depth $a_M + \gamma$ at best ask price plus the given value γ , and it is similar in Equations 8 and 9.

Now, we can define the kinetic energy and momentum of an order akin the physical kinetic energy and momentum of an object. That is, the product of half its size and squared velocity $\frac{1}{2}s \cdot v^2$ for kinetic energy, and the product of its size and velocity $s \cdot v$ for momentum. Thus, to calculate the kinetic energy and momentum of submitted market and limit orders we shall use equations (1) and (3) for velocity, and for cancelled limit orders – equations (2) and (4).

Recall the denotation of $N_\gamma(t)$ for the total number of limit orders at depth γ of the LOB at time t . We then calculate the cumulative sum of kinetic energy and momentum respectively at time t by aggregating the kinetic energy and momentum resulting from all order activities occurred in the order book depths γ ranging from the lower bound to the upper bound of the active area. Specifically, the kinetic energy of the order book at time T is

$$KE(T) = \sum_{t \in (0, T]} \sum_{\gamma^* = b_M - \alpha}^{a_M + \alpha} \left(\sum_{i=1}^{N_{\gamma^*}(t)} \frac{1}{2} s_i \cdot v_i^2 + \sum_{j=1}^{N_{\gamma^*}^c(t)} \frac{1}{2} s_j^c \cdot v_j^{c2} \right) \quad (8)$$

and the momentum of the order book at time T is

$$P(T) = \sum_{t \in (0, T]} \sum_{\gamma^* = b_M - \alpha}^{a_M + \alpha} \left(\sum_{i=1}^{N_{\gamma^*}(t)} s_i \cdot v_i + \sum_{j=1}^{N_{\gamma^*}^c(t)} s_j^c \cdot v_j^c \right) \quad (9)$$

where s_i and v_i are the order size and velocity of the i^{th} submitted limit order quoted at this depth, while s_j^c and v_j^c are the order size and velocity of the j^{th} cancelled limit order quoted at this depth.

Importantly, our approach models the order book as a complex system, and emphasises the vicinity surrounding the bid-ask spread, where order activities exhibit the highest degree of activity, and assigns greater weight to orders quoted in closer proximity to the bid-ask spread as they exert a more substantial influence on the bid-ask spread.

4.2. Calculation of Active Depth

This section explains the methodology used to calculate the Active Depth, a crucial component in the proposed Physical Model. We will present the details how to establish the definition and calculation on the upper and lower

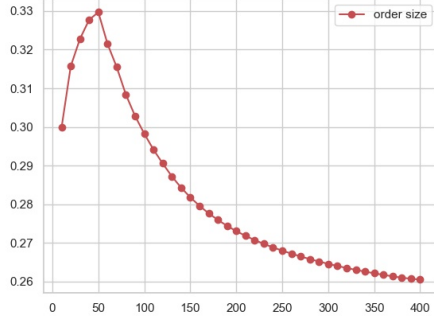
boundaries of the Active Depth. In Section 4, we introduced the concept of the active area, defined by the order book depth α , wherein the majority of order activity takes place. This section will focus on the computational methodology for deriving this pivotal depth and elaborate on the empirical analysis supporting its efficacy.

To extract this depth empirically and determine the active area α , we investigate the interconnection between the order activities around to the bid-ask spread and their response subjected to movements in the match price, in terms of the order cancellation or submission. Our approach employs a moving window to compute the change in the Reacted Orders Volume $\Delta V(\gamma)$ within different order book depth γ based on equation 7. Concurrently, we assess the corresponding distance traversed by the match price $v^M(t)$ within this identical time frame.

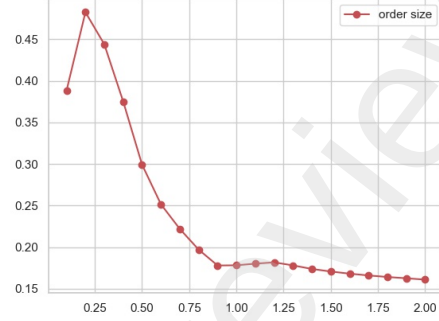
Our analysis further extends to study the cross-correlation function $\text{Corr}[v^M(t), \Delta V(\gamma)]$ to investigate the relationship between changes in the match price and alterations in the total volume of orders reacted at depth γ . The graphical representation of this relationship is presented in Figure 3, where the time range of the data investigated were 15:00-16:00 28/11/2022 for BTC/USD pair, and 02:00-03:00 12/05/2022 for LUNA/USD pair.

Specifically, to compute the cross-correlation coefficient $\text{Corr}[v^M(t), \Delta V(\gamma)]$ across order book depths, we proceed by applying a subtle methodology for each depth γ^* . First, we establish a moving window with a fixed step size. Within each window, we aggregate the total size of orders submitted and cancelled (Reacted Order Volume), resulting in an order size count that captures the volume of the responsive order activities at that depth. Simultaneously, we calculate the price velocity over each step, defined as the difference in the match price between the end of the previous window and the end of the current window. By constructing a series of Reacted Order Volume and corresponding price velocities, we then calculate the correlation between these two series, yielding the cross-correlation coefficient $\text{Corr}[v^M(t), \Delta V(\gamma^*)]$ for each depth γ^* . This approach allows us to quantify the relationship between match price velocity and the order activities corresponded to the price change across varying depths in the order book.

A close examination of Figure 3a reveals the correlation coefficient peaking at depth \$50 for the BTC/USD market. In contrast, for the LUNA/USD market, the maximum correlation is observed at a considerably shallower depth of \$0.2, as illustrated in Figure 3b. This is due to the different proportions of tick size (minimum price increment unit) to the price of the



(a) BTC/USD market



(b) LUNA/USD market

Figure 3: Correlation analysis of the match price and the change in Reacted Orders Volume across order book depth (X-axis: order book depth, Y-axis: correlation coefficient)

694 respective underlying asset. For Bitcoin, this ratio is approximately \$0.01 to
 695 about \$16,000, whereas for LUNA, it is significantly larger, at \$0.01 to about
 696 \$2. These graphical illustrations portray a quantifiable correlation between
 697 fluctuations in the match price and corresponding changes in the Reacted
 698 Orders Volume, as a function of the order book depth.

699 This characteristic pattern demonstrates a collective trend following be-
 700 haviour of orders, thereby substantiating our active area concept. In partic-
 701 ular, according to the experiments based on different windows sizes ranging
 702 from 1 second to 360 seconds, at an order book depth of \$50 for Bitcoin and
 703 \$0.2 for LUNA, the high-frequency traders exhibit a pronounced tendency
 704 to modify the largest volume of order in response to the movement of the
 705 match price.

706 Now we can define the Active Depth α by:

$$\alpha = \arg \max_{\gamma} \text{Corr}[v^M(t), \Delta V(\gamma)] \quad (10)$$

707 This empirical evidence amplifies the pivotal role of the active area in the
 708 financial market dynamics and underscores its utility in assessing the market
 709 microstructure. It offers a valuable lens through which to understand and
 710 anticipate the behaviour of high-frequency traders. The findings contribute
 711 to the broader understanding of market liquidity and depth, and could have
 712 important implications for market participants and regulators alike.

713 5. Empirical Findings

714 In this section we present the results of applying the Physical Model to
 715 the datasets of Bitcoin and LUNA. We report the outcomes of their analysis
 716 and interpret these findings, highlighting the ability of the Physical Model
 717 to describe the order book dynamics and market behaviour.

718 In the realm of physical systems, examining the motion of a single particle
 719 is infeasible. However, when considering a vast number of particles within a
 720 complex system, statistical rules governing their collective behaviour and mo-
 721 tion may be discerned. A parallel can be drawn with an order book dynam-
 722 ics, where individual traders place orders based on their supply and demand
 723 needs, resulting in irregular market oscillations and noise. Nevertheless, when
 724 aggregating all order records and information, emergent collective behaviours
 725 become evident. By observing these collective behaviours, insights into mar-
 726 ket behaviour can be gleaned, potentially unveiling the underlying factors
 727 driving the order book and price dynamics.

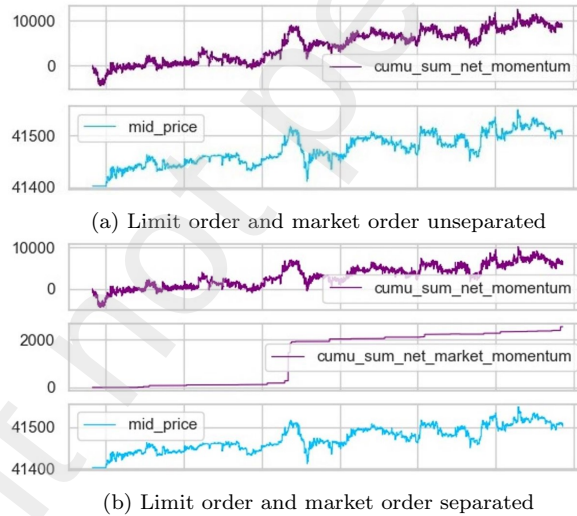


Figure 4: Measures in the active area (depth = \$100) for BTC/USD pair during 20/04/2022 23:00-00:00, with limit order and market order unseparated or separated

728 Thus, for the BTC/USD pair during the interval of 20/04/2022 23:00-
 729 00:00 and with an active area set to \$100, the cumulative sum of net momen-
 730 tum for all orders closely mirrors the midprice movement, as demonstrated
 731 in Figure 4a.

Examining the cumulative sum of net momentum separately for limit orders and market orders yields Figure 4b, where the first row represents the cumulative sum of net momentum for limit orders and the second row for market orders. We see a noticeable jump in the cumulative sum of net momentum for market orders, which is attributable to the placement of a substantial market buy order that does not adhere to the midprice trend.

Similar experiments on the LUNA/USD pair during 11/05/2022 16:00-17:00, 18:00-19:00 and 19:00-20:00 with active area set to \$0.5 are illustrated in Figures 5a, 5b and 5c.

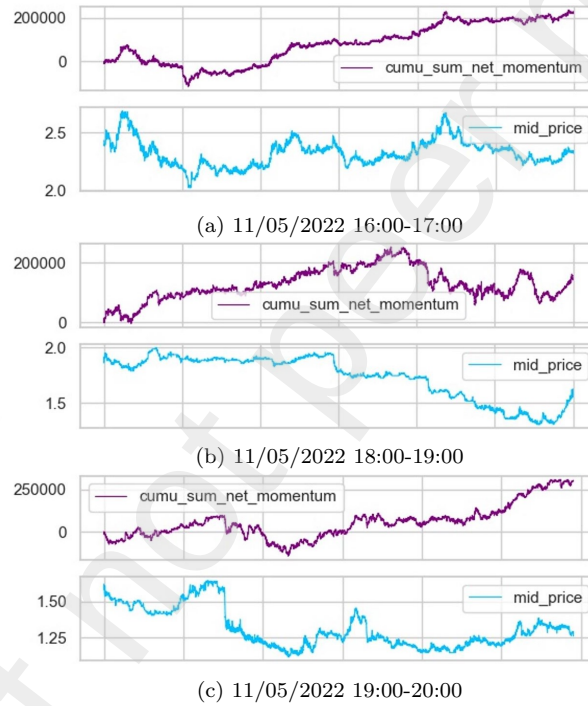


Figure 5: Measures for LUNA/USD during different periods

The midprice trend is followed albeit somehow more loosely than in the case of BTC/USD (e.g., the midprice downtrends in the middle of Figure 5a not reflected by the cumulative sum of net momentum). However, upon evaluating the cumulative sum of net momentum separately for limit orders and market orders in Figures 6a, 6b, and 6c, an intriguing market microstructure phenomenon emerges. Market orders consistently exhibit greater momentum than limit orders across all three time periods. Moreover, market orders

748 persistently attempt to drive the price upward (unsuccessfully), while limit
749 orders continuously sell.

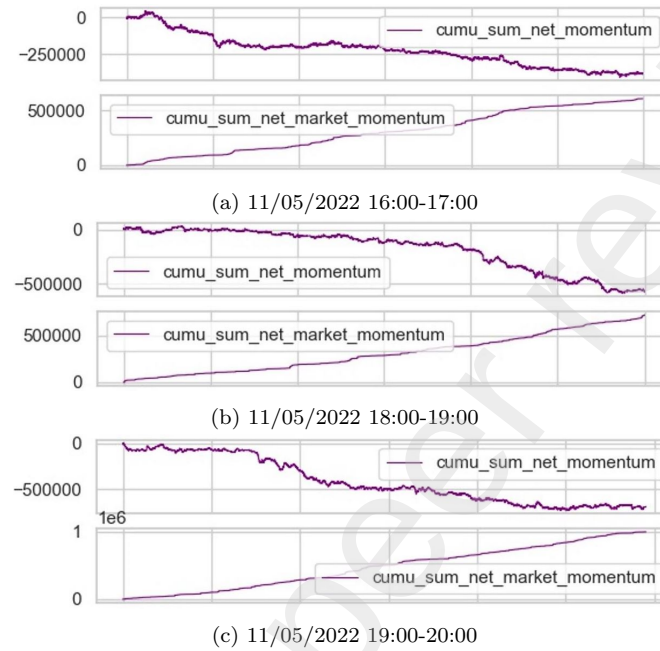


Figure 6: Measures in the active area (depth = \$0.5) for LUNA/USD pair during different periods, with limit order and market order separated

750 6. Volatility Forecasting: A Comparative Analysis of Kinetic En- 751 ergy and VPIN

752 In the realm of financial market analysis, predicting volatility is an ongoing
753 challenge and a critical component for both risk management and trading
754 strategies. Traditional volatility prediction models have often relied on
755 measures like historical volatility, implied volatility, GARCH models, and
756 more recently, VPIN (Volume-Synchronised Probability of Informed Trading).
757 However, we proposed a novel approach to predict short-term volatility
758 by employing the concept of Energy Change, inspired by physical systems.

759 The VPIN model discussed by Easley et al. (2012); Abad and Yagüe
760 (2012), is designed to measure the imbalanced volume of trades. It is based
761 on the idea that informed traders will trade more aggressively, leading to an
762 imbalance in the trade volume and hence, increased volatility. The model
763 has been popularly used for its ability to capture the arrival rate of informed

traders and its subsequent impact on volatility. However, the VPIN model primarily focuses on transaction data, and as a result, might not fully capture the dynamics of the limit order book.

In contrast, the kinetic energy measure proposed offers a fresh perspective by drawing an analogy between physical systems and financial markets. The Energy measure is calculated as half the sum of the squared order velocity in the defined Active Depth of the order book. This measure is expected to capture the degree of demand to produce a change for different order activities, thus evaluating a macroscopic view of the order book dynamics.

6.1. Introduction on the VPIN Method

The Volume-Synchronised Probability of Informed Trading (VPIN) represents a cutting-edge analytical tool, garnering increasing interest within the realm of market microstructure studies. This tool serves as a metric for the disequilibrium between informed and uninformed market trades. The cornerstone of VPIN lies in the hypothesis that informed traders leverage private information, thus provoking an imbalance in trading volumes.

To derive VPIN, one divides the trade volume into discrete segments, or 'buckets', and subsequently juxtaposes the volume of buyer-oriented trades to that of seller-initiated trades within each segment. Normalising the disparity between these two volumes by the total volume yields the VPIN. A high VPIN value signals a heightened likelihood of informed trading, indicative of a substantial disequilibrium between purchase and sale orders. In contrast, a low VPIN suggests a harmonious market characterised by equivalent levels of purchasing and selling.

In the sphere of financial markets, VPIN serves as an instantaneous indicator of market states, offering crucial insights into the degree of information asymmetry in the market. This asymmetry could presage periods of intense volatility. Hence, the VPIN may serve as an instrument for forecasting short-term market fluctuations, proving particularly beneficial for high-frequency traders and market makers.

Abrupt VPIN fluctuations, particularly sharp escalations, may serve as an early warning system for impending market volatility. Some empirical studies suggest that elevated VPIN values can anticipate price volatility driven by liquidity and market imbalances. Consequently, traders and financial institutions frequently employ VPIN as a risk management instrument. However, it is crucial to bear in mind that while VPIN is a valuable metric, it does carry its set of limitations. The VPIN model presumes that all informed trades will

instigate price variations, an assumption that may not always hold. Moreover, the selection of volume bucket size can significantly influence the VPIN output and there lacks a universal method choosing the optimal bucket size.

6.2. Historical Volatility

In the context of time series analysis, the Granger causality test proposed by Granger (1969) is a statistical hypothesis test for determining whether one time series is useful in forecasting another and provide evidence of predictive power.

The results of P-values in the Granger causality test on the 5-minute volatility against the VPIN time series lagging from 1 to 300 seconds for Bitcoin and LUNA are shown in Figure 7. The similar results on the 5-minute volatility against the kinetic energy are shown in Figure 8.

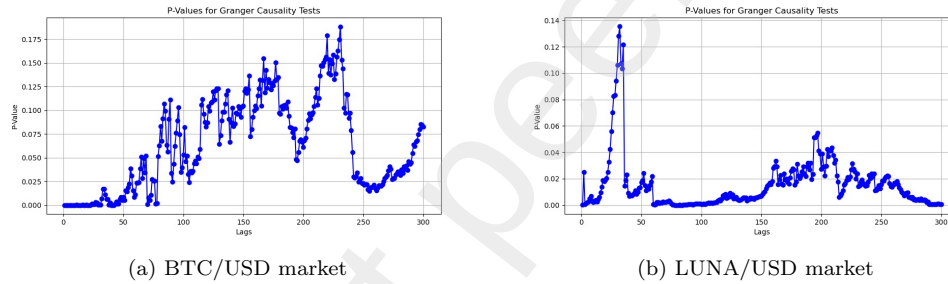


Figure 7: Granger causality test on the 5-minute volatility against the VPIN

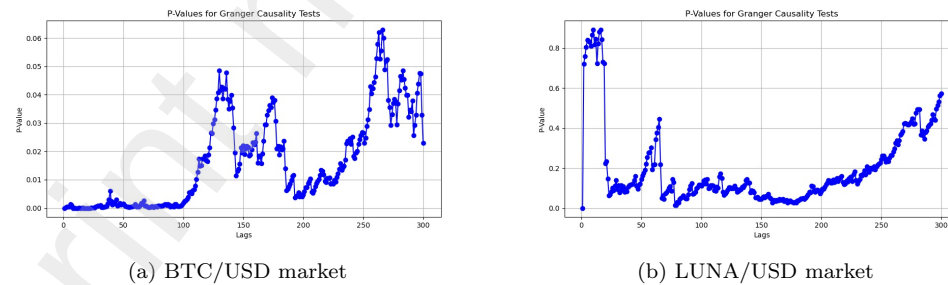


Figure 8: Granger causality test on the 5-minute volatility against the kinetic energy

A small p-value indicates strong evidence against the null hypothesis and suggests that the latter time series Granger-cause the former time series. The VPIN data performs the best in predicting the 5-minute volatility that is

816 about 0-50 seconds behinds for Bitcoin and about 55-150 seconds for LUNA.
817 And the kinetic energy performs the best predicting the 5-minute volatility
818 about 0-100 seconds behinds for Bitcoin while it acts terribly in predicting
819 the 5-minute volatility for LUNA.

820 In general, we can conclude that the VPIN of LUNA performs better
821 than its kinetic energy in predicting the 5-minute volatility. while the kinetic
822 energy of Bitcoin outperform its VPIN.

823 6.3. Local Volatility

824 As we are studying the high-frequency trading data, we adopt a different
825 method to calculate the instantaneous volatility. This new method seeks to
826 address a potential shortcoming of the traditional standard deviation calcu-
827 lation in the context of financial asset price data, particularly when the asset
828 price exhibits a trending behaviour.

829 The traditional calculation of volatility involves computing the standard
830 deviation of returns, which is based on the mean return. However, when the
831 asset price exhibits a trending behaviour, the mean return could be signifi-
832 cantly different from zero, and changing the length of the time window used
833 for the calculation could result in artificially high volatility values. This is
834 because more returns would be further away from the mean, leading to a
835 larger sum of squared deviations.

836 To address this issue, our Local Volatility measure computes the standard
837 deviation of the differences between consecutive asset prices, rather than their
838 returns. This provides a measure of how much the price changes from one
839 tick to the next, irrespective of the overall trend.

840 The Local Volatility for an asset with time series of match price P_t^M is
841 defined as:

$$\sigma_{LV} = \sqrt{\frac{\sum_{t=1}^T (p_t^M - p_{t-1}^M)^2}{T}} \quad (11)$$

842 where p_t^M represents the price at time t , and T is the total number of price
843 observations, the symbol σ_{LV} represents the Local Volatility.

844 This equation calculates the square of the differences between consecutive
845 prices $(p_t^M - p_{t-1}^M)^2$, sums them up, divides by the total number of observations
846 to calculate the mean squared difference, and finally takes the square root
847 to obtain the standard deviation of price changes. This Local Volatility
848 measure provides a more meaningful estimate of the instantaneous volatility

of an underlying asset, particularly when the asset price exhibits a trending behaviour.

Regarding to the calculated 30-second Local Volatility, we carried out the Granger causality test again, and the results of 30-second Local Volatility against the VPIN are shown in Figure 9 while that against the kinetic energy in Figure 10.

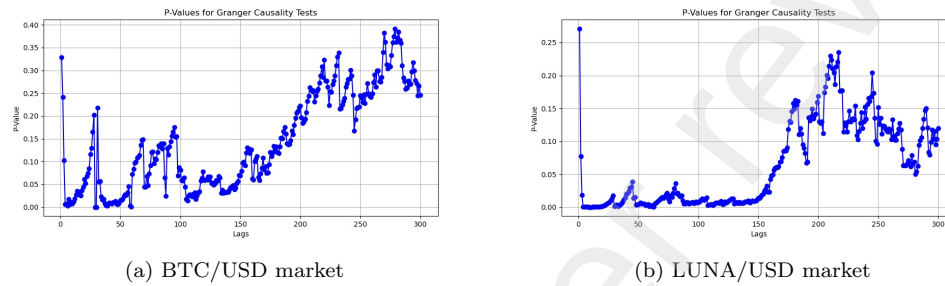


Figure 9: Granger causality test on the 30-second Local Volatility against the VPIN

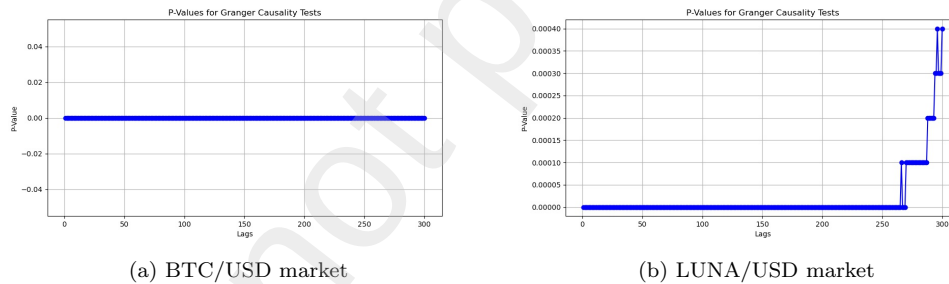


Figure 10: Granger causality test on the 30-second Local Volatility against the kinetic energy

It is apparent that the VPIN method has a worse performance in the instantaneous volatility data with higher sampling frequency. And these results prove that the kinetic energy owns significantly advantage compared to the VPIN method in predicting the instantaneous volatility in a high-frequency trading market. Hence, while VPIN is a powerful real-time measure of order flow toxicity, it does not fully capture limit-order dynamics at deeper levels of the LOB.

6.4. Comparative Analysis

The empirical analysis reveals that the kinetic energy measure outperforms the VPIN model in predicting short-term volatility for both Bitcoin and LUNA. This superior performance to the kinetic energy measure's ability to capture not only the effects of individual order activity but also the overall order book dynamics. Our model includes the LOB depth by computing the kinetic energy measure of each order activity according to the LOB depth of their quoted price so that their different impact on the order book can be analysed quantitatively, which are often overlooked by models focusing solely on transaction data.

The findings suggest that while the VPIN model has its merits, the kinetic energy measure provides a more encompassing view of order book dynamics and is thus a more effective tool for predicting short-term volatility. This highlights the potential benefits of integrating concepts from econophysics into financial markets.

These results underscore the importance of considering multiple facets of market data when predicting volatility. The kinetic energy measure's success points to the potential value of developing hybrid models that combine transaction data with other elements of the LOB. Such an approach could provide a more accurate and comprehensive understanding of the complex and rapidly changing dynamics that characterise financial markets.

In conclusion, the comparative analysis of the kinetic energy measure and the VPIN model represents a significant contribution to the literature on volatility prediction. It not only provides a novel perspective on volatility prediction but also sparks a potential shift in the methodological paradigm from purely econometric models to an interdisciplinary approach combining financial economics and econophysics. Further research exploring this interdisciplinary approach could open up new vistas for improving volatility prediction in financial markets.

7. Price Prediction: Momentum Versus Order Flow Imbalance

Forecasting price changes in financial markets forms the foundation of trading strategies and risk management paradigms. Conventional methodologies frequently depend on the Order Flow Imbalance (OFI), a metric that characterises shifts in supply and demand. We introduced a novel measure - the physical momentum, conceptually rooted in physical systems - for anticipating near-term price fluctuations.

The OFI has been widely recognised and utilised in microstructure studies, primarily to encapsulate the fundamental alterations in supply and demand dynamics within financial markets. The premise of the OFI model is underpinned by the belief that an augmentation in the best bid or size at the best bid is indicative of an escalation in demand, while a diminution signifies a contraction in demand. Inversely, a decrease in the best ask or an expansion in the size at the best ask points to a surge in supply, while an increment in the best ask or a contraction in size at the best ask signals a decrease in supply. Consequently, this metric measures the change in the supply or demand, even at a high-frequency level.

In contrast, the Momentum Change metric, as proposed in this study, takes its conceptual basis from physical systems, and seeks to encapsulate the 'momentum' of the order book system. It is calculated as the aggregated product of the velocity and the corresponding volume of each order. By incorporating the impact of each order on the price movement based on their LOB depth and quoted volume, this measure provides a more holistic view of the order book dynamics.

7.1. Introduction on the OFI

The OFI measure introduced by Cont et al. (2014) is a pivotal instrument in the domain of market microstructure, playing an instrumental role in assessing the swings in supply and demand dynamics. Conceived as an answer to the demand for a sophisticated understanding of these oscillations, the OFI measure is firmly rooted in empirical finance research and has been extensively employed as a tool for market examination.

In theoretical terms, the OFI measure is founded on the premise that the oscillations in the best bid and ask levels, in conjunction with the corresponding sizes, can act as an effective surrogate for the latent supply and demand changes in the market. In specific terms, an augmentation in the best bid or an augmentation in the size at the best bid is construed as a surge in demand, signaling an influx of purchasing interest in the market. Inversely, a diminution in these variables signifies a contraction in demand, indicating a shrinkage in buying interest.

On the supply facet, the OFI measure operates under an analogous rationale. A diminution in the best ask or an augmentation in the size at the best ask is perceived as a swell in supply, suggesting an increasing willingness of market participants to vend. In contrast, an augmentation in the best ask

934 or a diminution in size at the best ask indicates a contraction in supply,
 935 signifying a retrenchment in selling interest.

936 The OFI measure, therefore, acts as a vital metric that encapsulates high-
 937 frequency changes in market dynamics. By providing real-time insights into
 938 the state of supply and demand in the market, the OFI measure aids the
 939 creation of reactive and adaptive trading strategies. It enables market par-
 940 ticipants to forecast price movements and adapt their positions accordingly,
 941 thus mitigating risk and optimising returns.

942 7.2. The Prediction Power on Price Change

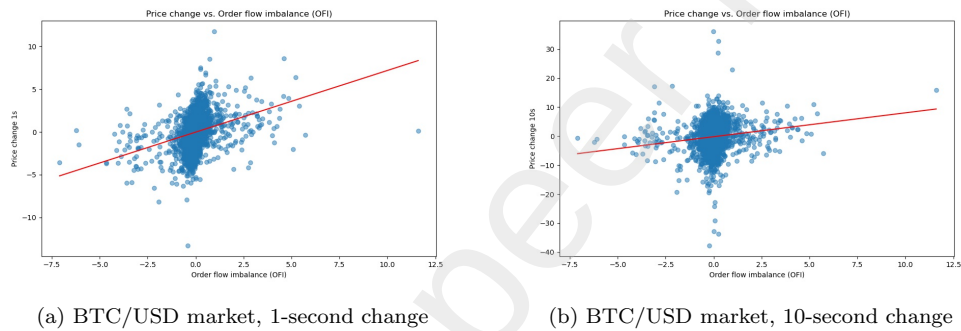


Figure 11: Regression analysis on the match price change against the OFI

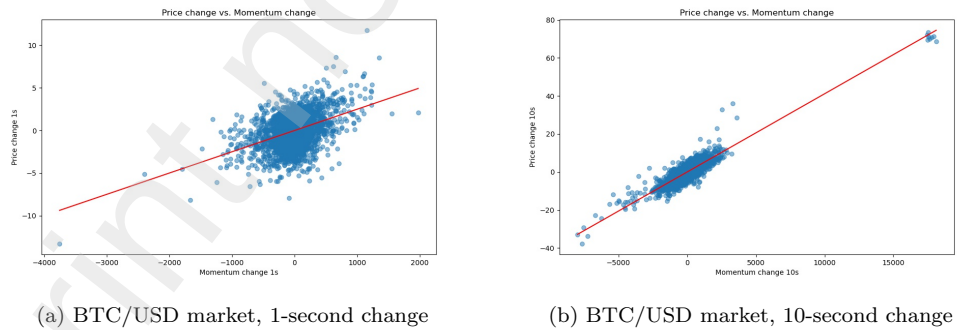
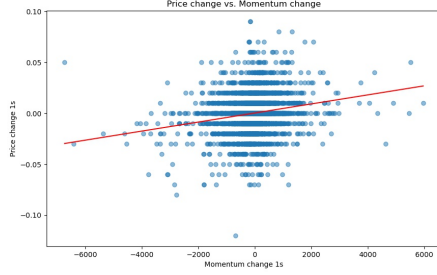
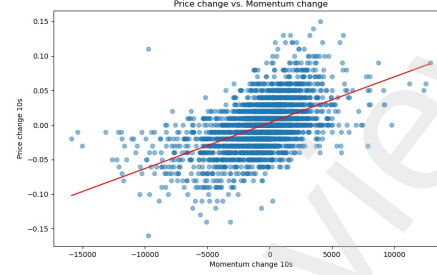


Figure 12: Regression analysis on the match price change against the momentum

943 This section provides a comprehensive empirical analysis of the predictive
 944 power of two different measures, namely the Order Flow Imbalance (OFI) and
 945 Momentum Change, in forecasting the price change of Bitcoin and LUNA



(a) LUNA/USD market, 1-second change



(b) LUNA/USD market, 10-second change

Figure 13: Regression analysis on the match price change against the momentum

946 over two distinct time horizons: one and ten seconds. For LUNA, due to
 947 unavailability of certain data points, the OFI measure could not be computed,
 948 and thus we only report the results for physical momentum change.

949 In order to evaluate the respective predictive abilities of trade imbalances
 950 and Order Flow Imbalances with regards to price fluctuations, we conduct a
 951 series of the following regressions:

$$\Delta P_k = \alpha_i + \beta_i OFI_k + \epsilon_k \quad (12)$$

952 for the OFI measure and

$$\Delta P_k = \alpha_i + \beta_i M_k + \epsilon_k \quad (13)$$

953 for the momentum measure.

954 These regressions are computed independently for each data subset (in-
 955 dexed by i). Each subset represents an one-hour order book data sampled
 956 by 1-second frequency, which typically comprises approximately 3600 data
 957 points (indexed by k), collected at a frequency of one second. The details for
 958 these regression analysis are listed in Table 2.

959 The OFI measure was regressed against price changes of Bitcoin over one-
 960 second and ten-second horizons according to Figure 11. The R-squared values
 961 were 0.094 and 0.019 for the one-second and ten-second horizons, respectively.
 962 These relatively low R-squared values suggest that the predictive power of
 963 the OFI measure is modest, but not negligible. The F-statistic results (373.0
 964 and 70.87) and their corresponding p-values (3.83e-79 and 5.45e-17) indicate
 965 that the OFI measure is statistically significant in predicting Bitcoin price
 966 changes. Meanwhile, the performance of the OFI on Bitcoin data is relatively

Table 2: The regression analysis

Dataset	R^2	F-statistic	Prob(F-statistic)	Omnibus	Prob(Omnibus)	Skew	Kurtosis	Durbin-Watson	Jarque-Bera (JB)	Prob(JB)
BTC/USD OFI vs ΔP 1s	0.094	373.0	3.83e-79	255.040	0.000	-0.139	5.829	2.472	1208.261	4.26e-263
BTC/USD OFI vs ΔP 10s	0.019	70.87	5.45e-17	655.211	0.000	-0.403	11.230	0.337	10204.282	0.00
BTC/USD ΔM vs ΔP 1s	0.168	723.6	2.33e-145	110.277	0.000	-0.128	4.296	2.653	261.177	1.93e-57
BTC/USD ΔM vs ΔP 10s	0.816	1.589e+04	0.00	747.131	0.000	0.737	9.455	0.970	6560.394	0.00
LUNA/USD ΔM vs ΔP 1s	0.036	137.0	4.26e-31	202.154	0.000	0.012	5.316	2.793	816.435	5.17e-178
LUNA/USD ΔM vs ΔP 10s	0.274	1375	1.43e-255	165.117	0.000	0.236	4.593	0.840	419.270	9.05e-92

weaker than the result presented by Cont et al. (2014) where the R-squared value is about 0.65 for NYSE TAQ stock data, which indicates that on the Bitcoin data the OFI is not as efficient as on the equities data.

The Momentum Change measure was regressed against price changes of Bitcoin and LUNA over one-second and ten-second horizons in Figure 12. For Bitcoin, the R-squared values were 0.168 and 0.816 for the one-second and ten-second horizons, respectively, suggesting a stronger predictive power than the OFI measure. The F-statistics (723.6 and 1.589e+04) and their corresponding p-values (2.33e-145 and 0.00) further support the significant role of Momentum Change in predicting Bitcoin price changes. The coefficients of the Momentum Change measure were positive and significant, suggesting that an increase in momentum would lead to a price increase in Bitcoin. On the other hand, this feature is not significant enough for the LUNA data in Figure 13.

Overall, our empirical analysis reveals that the Momentum Change measure has a stronger predictive ability for the price change of Bitcoin compared to the OFI measure. This suggests that the Momentum Change measure could be a more robust tool for short-term price prediction in the cryptocurrency market.

7.3. Comparative Analysis

The empirical study offers a comparative evaluation of these two metrics. For Bitcoin, it was discovered that the Momentum Change metric demonstrated superior performance than the OFI in forecasting price fluctuations over one-second and ten-second intervals. The R-squared values associated with the Momentum Change metric were significantly larger than those affiliated with the OFI metric, suggesting a more potent predictive capacity. The statistical significance of the Momentum Change metric, as evidenced by the F-statistics and their corresponding p-values, further accentuates its efficacy in forecasting Bitcoin price fluctuations.

Table 3: The fitted model

The fitted model	Coefficient	Standard error	t	$P \geq t $	[2.5%,	97.5%]
α_1	-0.0215	0.027	-0.787	0.431	-0.075	0.032
β_1	0.7208	0.037	19.313	0.000	0.648	0.794
α_2	-0.2008	0.072	-2.803	0.005	-0.341	-0.060
β_2	0.8237	0.098	8.418	0.000	0.632	1.016
α_3	-0.0070	0.026	-0.269	0.788	-0.058	0.044
β_3	0.0025	9.28e-05	26.900	0.000	0.002	0.003
α_4	0.0005	0.039	0.013	0.989	-0.076	0.077
β_4	0.0041	3.26e-05	126.061	0.000	0.004	0.004
α_5	0.0002	0.000	0.729	0.466	-0.000	0.001
β_5	4.441e-06	3.79e-07	11.706	0.000	3.7e-06	5.18e-06
α_6	0.0034	0.000	7.305	0.000	0.003	0.004
β_6	6.637e-06	1.79e-07	37.075	0.000	6.29e-06	6.99e-06

In the case of LUNA, due to data limitations, only the Momentum Change metric could be computed. However, the results indicated a substantial predictive capability of the Momentum Change metric for LUNA prices, especially over a ten-second interval.

The results imply that while the OFI metric provides critical insights into the dynamics of supply and demand, the Momentum Change metric, by amalgamating price and volume fluctuations, offers a more comprehensive understanding of market dynamics. This underscores the potential advantages of integrating concepts from the physical sciences into financial market analysis. The superior performance of the momentum measure over the OFI implies potential limitations of the OFI in forecasting price fluctuations in cryptocurrency data, particularly when compared with stock data. This underscores the distinct characteristics of cryptocurrency markets and the necessity for further experiments of the Physical Model on the other asset classes.

The comparative empirical study between the Momentum Change metric and the OFI model constitutes a notable addition to the literature on price change prediction. It not only introduces a fresh perspective but also indicates a paradigm shift from traditional econometric models to an interdisciplinary approach that merges financial economics and physical sciences. Further investigation into this interdisciplinary approach could potentially facilitate advancements in price change prediction within financial markets.

8. Benchmarking the Physical Model: Insights Into Predictive Performance

Based on the work in Sirignano (2019), the Deep Long Short Term Memory (LSTM) model designed by Sirignano and Cont (2019) was constituted by three layers of LSTM units, succeeded by a fully-connected layer of rectified linear units (ReLUs). The subsequent application of a softmax activation function facilitated the generation of a probability distribution for the future price movement. Sirignano and Cont (2019) compared the performance of the Deep LSTM model against the other linear models, especially Vector Autoregressive (VAR) model. The empirical results revealed that the Deep LSTM model transcended these counterparts, demonstrating its capability to capture non-linear relationships in the LOB data which other models might miss.

The DeepLOB model, further proposed in Zhang et al. (2019), introduces a deep learning framework for predicting short-term price movements using limit order book (LOB) data. It combines convolutional neural networks (CNNs) to extract spatial features from LOB snapshots with long short-term memory (LSTM) networks to capture temporal dependencies. This hybrid architecture enables effective learning from high-dimensional, noisy market data. DeepLOB demonstrates strong predictive performance across multiple instruments and maintains generalisability to unseen stocks, making it a robust tool for financial forecasting.

We replicated the aforementioned two models as baseline to test the capability of our Physical Model. Our dataset comprises snapshots of the LOB, encapsulating price and volume data for the 10 highest bid and 10 lowest ask orders. The snapshots data was reconstructed from the Coinbase exchange for Bitcoin/USD pair during 2023/09/16 02:00 - 2023/09/16 14:00. Note that we did not test the Deep LSTM model on LUNA/USD pair during flash crash due to the lack of corresponding snapshots data.

Each observation or timestamp in the raw data thus constitutes 40 features: 10 bid prices, 10 bid volumes, 10 ask prices, and 10 ask volumes. We computed the physical measure with the Physical Model and synchronised them to the original snapshot data. The snapshot dataset contained 2,018,900 LOB snapshots in the span of 12 hours, and the 80% of them are used as training set while the left for the test set. The physical measures are computed from the Level-3 order book data with length 9,839,922 during the same time span. We also carried out further experiments by incorporating

1055 the physical measures including Market Momentum and Market Energy as
 1056 the training data. The classification of training set and test set is depicted
 1057 in Figure 14, where the data labelled with blue dotted line is the training set
 1058 and red for the test set.

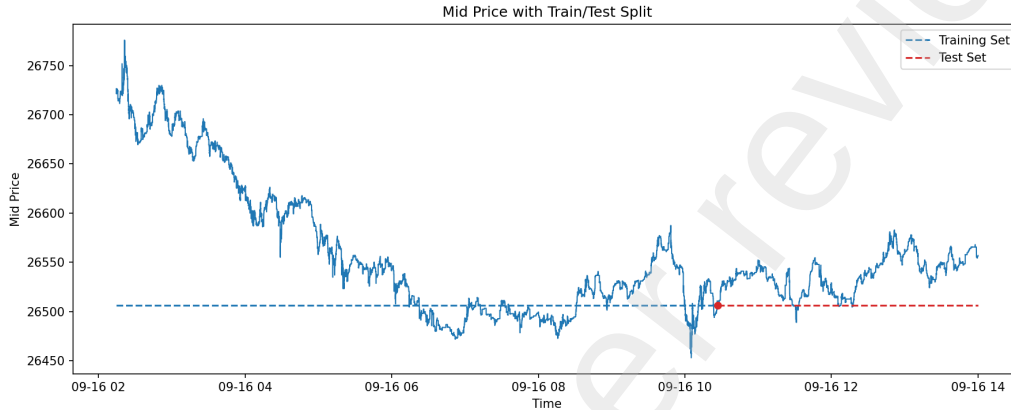


Figure 14: Training/Test partition on the dataset

1059 Before feeding the data into the LSTM, we subjected it to rigorous prepro-
 1060 cessing. We standardised the dataset ensuring zero mean and unit variance
 1061 for each feature. This scaling is imperative for neural networks to ensure
 1062 all input features are treated with equal importance and to aid in model
 1063 convergence.

1064 8.1. Benchmark Models

1065 **The Deep LSTM model.** Sirignano and Cont (2019) utilised a deep
 1066 feedforward neural network, where the input was several layers of the limit
 1067 order book (i.e., levels of bids and asks, quantities at each level, etc.), and the
 1068 output was the future price change after a certain fixed time horizon. They
 1069 used a large dataset to train the neural network, optimising the model's
 1070 weights using backpropagation. The model was trained to minimise the
 1071 difference between its predicted price changes and the actual price changes
 1072 observed in the data.

1073 Instead of handcrafting features as in traditional finance models, the Deep
 1074 LSTM model was trained to directly learn from the order book data. This

automatic feature extraction capability is crucial for understanding complex structures and patterns in financial data.

Sirignano and Cont (2019) compared the performance of the Deep LSTM model with other benchmark models. The Deep LSTM outperformed these models, demonstrating its capability to capture non-linear relationships in the LOB data which other models might miss. Apart from its predictive power, the model's importance lies in its ability to shed light on the microstructure of financial markets. The patterns and features automatically learned by the Deep LSTM model can be interpreted to gain insights into the underlying mechanisms of price formation in financial markets.

In essence, their study showcases the power of deep learning in capturing and understanding the complex dynamics of financial markets, particularly when it comes to the microstructure of how prices are formed. It also emphasises the potential of deep learning to unveil universal features of price formation.

The model is designed to accommodate sequences of fixed length $T = 10$, each embedding 40 features, reflective of the bid-ask dynamics in our dataset. The network consists of a stack of three LSTM layers, each with 32 hidden units. These layers use the hyperbolic tangent (`tanh`) activation for hidden states and the hard-sigmoid function for recurrent activations. Dropout with a rate of 0.2 follows each LSTM layer to prevent overfitting. The LSTM layers are configured with `return_sequences=True` for the first two layers and `return_sequences=False` for the third, effectively summarising the sequence into a final hidden representation. The kernel and recurrent weight initialisers follow the Glorot Uniform and Orthogonal schemes, respectively, ensuring stable training. This deep configuration enables the model to capture hierarchical temporal patterns across the order book dynamics. The structure of the Deep LSTM layers is demonstrated as follows:

- Window length: Each training example comprises a sequence of the last $T = 10$ snapshots.
- Feature dimension: $F = 40$ raw depth-of-book features per snapshot, giving an input tensor of shape $(T, F) = (10, 40)$.
- Target: The continuous mid-price change $\Delta p_{t+0.5s}$ observed half a second after the final snapshot in each window.
- Input layer: `Input(10, 40)`.

- 1110 • Recurrent layers: A stack of three LSTM(32) layers. The first two lay-
1111 ers use `return_sequences=True`, while the final uses `return_sequences=False`.
- 1112 • Regularisation: Dropout(0.2) applied after each LSTM layer.
- 1113 • Dense layer: Dense(32, ReLU) adds a non-linear transformation before
1114 output.
- 1115 • Output layer: Dense(1, linear) predicts the continuous price change.

1116 The model employs a fully connected dense layer with a linear activation
1117 function for regression output. The chosen optimiser for the model was
1118 Adam with a learning rate of 10^{-4} , known for its efficient handling of sparse
1119 gradients and adaptability. The primary metric for model evaluation was the
1120 mean squared error (MSE), alongside directional accuracy for classification
1121 insight.

1122 The model was trained over 30 epochs with early stopping patience set to
1123 10. The batch size, a crucial hyperparameter, was set at 256, indicating the
1124 model processes 256 sequences simultaneously in each iteration to balance
1125 convergence stability and memory efficiency. This configuration supports
1126 effective learning of short-term price movements driven by order book dy-
1127 namics.

1128 **The DeepLOB model.** This subsection provides an academically rig-
1129 orous exposition of the DeepLOB model, a convolutional - recurrent neural
1130 network expressly designed for limit order book (LOB) data.

1131 The original formulation by Zhang et al. (2019) couples spatial convo-
1132 lutions with temporal recurrent layers to capture local depth-of-book pat-
1133 terns and their evolution across time. In contrast with earlier LSTM-only
1134 approaches, DeepLOB first learns hierarchical representations of the spa-
1135 tial structure of the LOB (e.g. bid-ask levels, volume imbalances) and then
1136 models their temporal dynamics, thereby extracting richer features for price-
1137 movement prediction.

1138 Similar to the Deep LSTM examined previously, DeepLOB eschews hand-
1139 crafted features in favour of end-to-end learning. However, by incorporating
1140 convolutional kernels of varying receptive fields together with an Inception-
1141 style module, it is able to detect short-, medium- and long-range level interac-
1142 tions that a pure recurrent model might overlook. Empirical results in Zhang
1143 et al. (2019) demonstrate that this hybrid architecture outperforms both tra-
1144 ditional statistical models and purely recurrent deep networks, highlighting

1145 the explanatory power of convolutional feature maps for market microstruc-
1146 ture. The architecture of our implemented DeepLOB model is as follows:

- 1147 • Input layer: A tensor of shape $(T, N_F, 1)$ where $T = 10$ time steps
1148 and $N_F = 40$ depth-wise features (yielding $10 \times 40 = 400$ raw LOB
1149 features).
- 1150 • Convolutional block 1:
 - 1151 – Conv2D(16, (1, 2), stride = (1, 2)) with LeakyReLU;
 - 1152 – two Conv2D(16, (4, 1)) layers with **same** padding and LeakyReLU
1153 activations.

1154 Reduces the feature-axis width from 40 to 20 while extracting local
1155 level interactions.
- 1156 • Convolutional block 2: Repeats the above pattern, compressing the
1157 width from 20 to 10 and refining spatial features (Conv2D(16, (1, 2))
1158 then two Conv2D(16, (4, 1))).
- 1159 • Feature-collapse block:
 - 1160 – Conv2D(16, (1, 10)) to collapse width $10 \rightarrow 1$, followed by two
1161 Conv2D(16, (4, 1)) layers (all with LeakyReLU).

1162 Produces a $(T, 1, 16)$ representation.
- 1163 • Inception-style module: Three parallel branches:
 - 1164 – 1×1 then 3×1 convolutions with 32 filters,
 - 1165 – 1×1 then 5×1 convolutions with 32 filters,
 - 1166 – 3×1 max-pool followed by a 1×1 convolution with 32 filters,

1167 each using LeakyReLU and concatenated to yield $(T, 1, 96)$.
- 1168 • Reshape & Dropout: The tensor is reshaped to $(T, 96)$ and a dropout
1169 layer ($p = 0.2$) is applied to mitigate overfitting.
- 1170 • Temporal layer: A single LSTM(16) processes the ten time steps, sum-
1171 marising temporal dependencies.
- 1172 • Output layer: A fully connected Dense(1) with linear activation outputs
1173 the predicted mid-price change Δp .

Model parameters and training details. The implementation adopts the Adam optimiser with a learning rate of 10^{-4} and a mean-squared-error loss, reflecting the continuous nature of the prediction target. Training is performed for 30 epochs with early stopping patience set to 10, using a batch size of 256 on 80% of the chronologically ordered data (with 10% of that subset reserved for validation). Empirical tuning confirmed that using 16 convolutional filters and 16 LSTM units strikes a good balance between model capacity and overfitting risk.

In summary, DeepLOB extends pure recurrent frameworks by embedding convolutional inductive biases that account for the hierarchical structure of order books, delivering state-of-the-art predictive performance and offering interpretable insights into level interactions within high-frequency markets.

The LOB-Transformer model. This subsection presents the Limit Order Book (LOB) Transformer model - a hybrid patch-based Vision Transformer (ViT) augmented with an LSTM read-out layer - tailored for high-frequency market data. Inspired by the patch-embedding strategy of the original Transformer architecture Vaswani et al. (2017), the model first tokenises the two-dimensional LOB snapshot into non-overlapping patches, projects them into an embedding space, and then applies stacked self-attention blocks to learn contextual interactions before a recurrent layer captures longer-horizon temporal dependencies.

Unlike convolutional approaches such as DeepLOB, the Transformer relies on global self-attention, permitting each patch token to attend to every other, thereby modelling higher-order cross-level relationships without inductive biases about locality. The subsequent LSTM summarises the sequence of attended patch embeddings, furnishing an expressive representation for mid-price change regression. We implemented an LOB-Transformer model with the following architecture:

- Input tensor: $(20, 20, 1)$, obtained by reshaping each $(10, 40)$ LOB window into a “pseudo-image” of height 20 and width 20 (so that $20 \times 20 = 400$ raw features).
- Patch extraction: A **Patches** layer splits the (20×20) image into $4 \times 4 = 16$ non-overlapping patches of size (5×5) , flattening each patch into a vector of length $5 \cdot 5 \cdot 1 = 25$.
- Patch encoder: Each patch vector is linearly projected to a $d = 32$ -

1209 dimensional embedding and augmented with learned positional em-
1210 beddings, producing a sequence of length 16 and dimension 32.

- 1211 • Transformer encoder (repeated twice):
 - 1212 – Layer normalisation;
 - 1213 – multi-head self-attention with 2 heads (`key_dim=32`) and dropout
 - 1214 rate 0.1;
 - 1215 – residual addition;
 - 1216 – MLP block with hidden dimensions [64, 32], ReLU activations,
 - 1217 and dropout 0.1;
 - 1218 – second residual addition.

1219 These operations yield contextualised patch embeddings of shape (16, 32).

- 1220 • Representation layer: Layer normalisation followed by dropout $p = 0.5$
1221 for regularisation.
- 1222 • Temporal aggregation: A single LSTM(64) (returning sequences) pro-
1223 cesses the 16 patch embeddings; its final hidden state is passed through
1224 batch normalisation.
- 1225 • Dense head: A Dense(64, ReLU) layer with batch normalisation and
1226 dropout 0.2 adds a non-linear transformation before the output.
- 1227 • Output layer: A Dense(1) with linear activation outputs the predicted
1228 mid-price change Δp .

1229 *Model parameters and training details..* The model is instantiated with `projection_dim=`
1230 32, two Transformer layers, two attention heads, and an LSTM of 64 units,
1231 followed by a dense layer of 64 units. The Adam optimiser with learning
1232 rate 3×10^{-6} minimises the mean-squared-error loss. Training proceeds for
1233 up to 50 epochs with early stopping patience set to 10, using a batch size of
1234 1024 on the earliest 80% of samples, reserving 10% of that subset for valida-
1235 tion and 20% for out-of-sample evaluation. Dropout rates of 0.1 within the
1236 Transformer, 0.5 after the Transformer, and 0.2 in the dense head mitigate
1237 overfitting given the limited number of epochs.

In essence, the LOB-Transformer model integrates global self-attention with recurrent temporal modelling, providing a flexible framework for capturing both spatial depth-of-book interactions and their evolution across time in high-frequency trading environments.

8.1.1. The Prediction Accuracy

Exploratory Analysis of the Experimental Set-up

This study assesses whether physically-motivated order book features can enhance short-horizon price-direction forecasting. Multiple experiments were carried out:

- Physics-based baseline: The Physical Model converts full Level-3 order book data into the cumulation of four measures - kinetic energy of order submission, kinetic energy of order cancellation, upward momentum, and downward momentum computed from the methods in Section 4 - without any parameter training. When evaluated on an event-driven test set the model achieved a directional accuracy of 64.44 % at a 0.5 s horizon and 61.88 % at a 1 s horizon.
- Deep-learning baselines: Three architectures commonly used for LOB forecasting were implemented: Deep LSTM by Sirignano and Cont (2019), DeepLOB by Zhang et al. (2019), and a proprietary LOB-Transformer. Each network was trained (i) on Level-2 snapshot data only, (ii) on the physical measures only, and (iii) on an augmented input consisting of snapshots plus the four physical measures.

Data alignment: Level-3 data arrive at a much higher frequency than the snapshots supplied. For each ticker, the cumulative physical measures were therefore merged into the snapshot stream that selects the most recent physical observation at or before each snapshot time stamp. The resulting dataframe provides a single, temporally aligned input row for every snapshot.

Accuracy metrics: The model performance were estimated using the directional predicting accuracy function given by Sirignano and Cont (2019) in equation 14, as the models were trained to forecast the direction of the next price change. Note that the number of correct prediction is calculated from the total correct forecasts of the model on the direction of price changes every time an event of price change occurs, where the datapoints are not sampled equally but are event-driven.

$$DirectionalAccuracy = \frac{\text{Number of correct predictions}}{\text{Total number of price changes}} \times 100\%. \quad (14)$$

In our study, two notions of different accuracies are reported:

- Sklearn Accuracy is the standard **sklearn** score: the proportion of samples for which the predicted sign of the price change matches the true sign, counting every sample, including those in which either value is exactly zero.
- Directional Accuracy is better for evaluating high-frequency predictions decreasing the impact of market microstructure noise: it ignores any observation where the true or predicted price change is zero and then measures the fraction of sign agreements among the remaining, strictly non-zero cases. Consequently this figure can diverge from the conventional accuracy, especially when the test set contains many negligible price moves.

In general, during training and testing the following statistics were recorded: directional accuracy, sklearn accuracy, precision, recall, F_1 , and ROC AUC.

Experimental Analysis

All deep models were optimised with cross-entropy loss on binarised direction labels. Test-set results for every architecture and input variant are listed in Table 4 for all models.

In addition to the machine-learning experiments, we also evaluated a simple physical momentum predictor that uses only the recent change in cumulative net momentum to forecast the sign of the mid-price change, without any trainable parameters. Specifically, for each timestamp t , we compute

$$\Delta P_{\text{past}} = P(t) - P(t - 0.1 \text{ s}),$$

where $P(\cdot)$ denotes the cumulative sum of net momentum. We then compare ΔP_{past} with the actual mid-price change over the next horizon $\Delta p_{\text{future}} = price(t + h) - price(t)$ (with $h = 0.5 \text{ s}$ or 1 s). A correct directional prediction is counted whenever ΔP_{past} and Δp_{future} share the same sign (both positive or both negative). Because this heuristic involves no fitted parameters, standard metrics that require model-based probability estimates

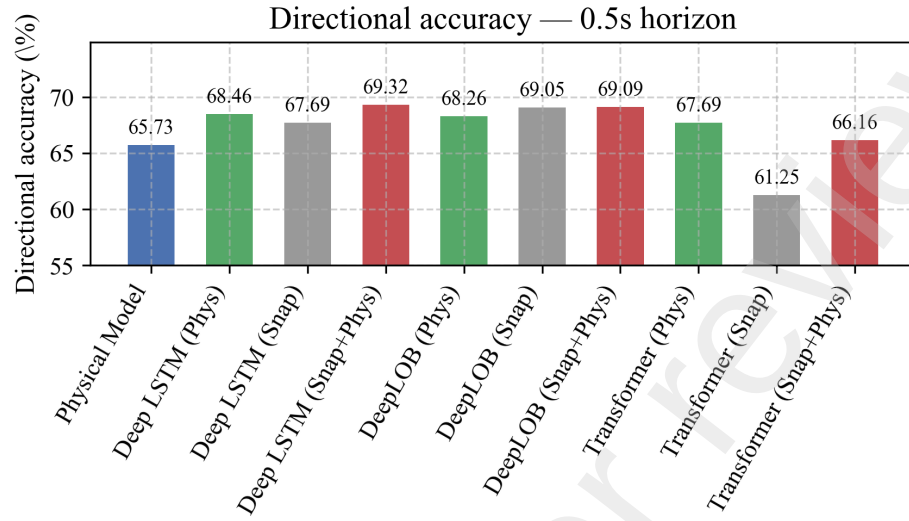
(such as sklearn accuracy, precision, recall, F1, or ROC AUC) are not defined. Instead, we report only the resulting directional accuracy-termed directional accuracy - which is simply the fraction of timestamps for which $\text{sign}(\Delta P_{\text{past}}) = \text{sign}(\Delta \text{price}_{\text{future}})$. Consequently, metrics such as accuracy, precision, recall, F1, and ROC AUC pertain only to models that involve learned parameters; they are omitted for the physical-momentum baseline.

Although the neural networks attain the highest headline scores, it is noteworthy that the Physical Model, which requires no training, still achieves accuracies of 65.7% (0.5 s horizon) and 64.0% (1 s horizon). In other words, a purely mechanistic description of order book dynamics already delivers a small but statistically significant forecasting edge without the computational overhead of deep learning. These results establish the Physical Model as a robust baseline.

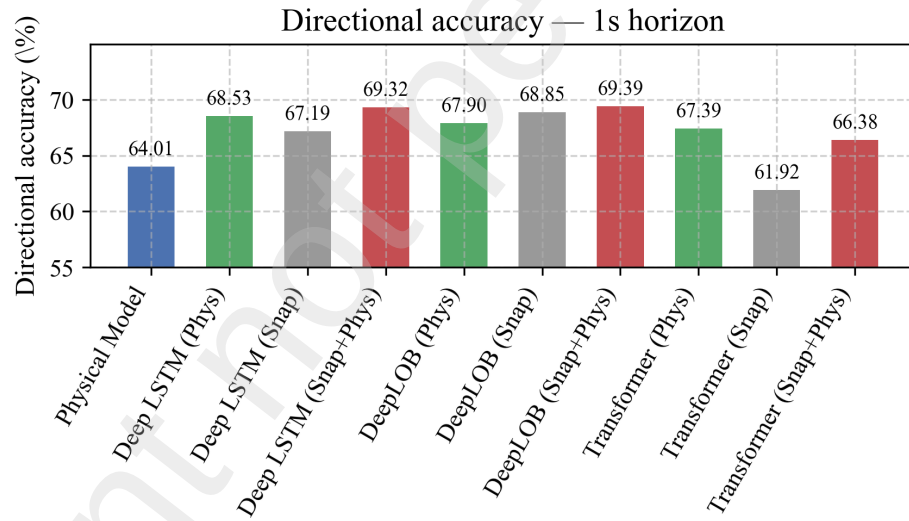
The experimental results from all settings are listed in Table 4, while the comparisons on the directional accuracy metric are demonstrated in Figure 15.

The experiments with predicting time horizon of 0.5 s are shown in Figure 15a, where physics-derived features (orange) consistently outperform snapshot - only baselines (grey), while deep networks fed solely with physical measures (green) recover most of the benefit without any depth-of-book images. The untrained Physical Model (blue) remains a strong lightweight benchmark.

As for predicting time horizon of 1 s compared in Figure 15b, the performance patterns mirror the 0.5 s horizon: combining physics-derived features with LOB snapshots yields the best results, whereas physics-only deep models still close much of the gap, confirming the predictive power embedded in the four physical measures alone.



(a) Directional accuracy at a 0.5 s prediction horizon



(b) Directional accuracy at a 1 s prediction horizon

Figure 15: Comparisons on the directional accuracy across different models, experimental settings and datasets

Table 4: Comparison of evaluation metrics across models and prediction horizons

Model & Horizon	Directional Accuracy	Sklearn Accuracy	Precision	Recall	F1	ROC AUC
Physical Momentum - resampled to snapshot time (0.5 s)	0.657301	-	-	-	-	-
Physical Momentum - resampled to snapshot time (1 s)	0.640118	-	-	-	-	-
Deep LSTM - physical measures only (0.5 s)	0.684597	0.643175	0.590833	0.674087	0.629720	0.692204
Deep LSTM - physical measures only (1 s)	0.685300	0.649411	0.610430	0.702113	0.653069	0.700474
DeepLOB - physical measures only (0.5 s)	0.682560	0.651162	0.608853	0.629281	0.618898	0.684500
DeepLOB - physical measures only (1 s)	0.679050	0.651779	0.625133	0.647130	0.635941	0.691433
LOB-Transformer - physical measures only (0.5 s)	0.676852	0.640404	0.590698	0.654895	0.621142	0.677958
LOB-Transformer - physical measures only (1 s)	0.673885	0.646862	0.619856	0.642864	0.631150	0.688431
Deep LSTM - snapshot only (0.5 s)	0.676897	0.636452	0.583362	0.672958	0.624965	0.681198
Deep LSTM - snapshot data only (1 s)	0.671923	0.640401	0.604406	0.679807	0.639893	0.683927
DeepLOB - snapshot only (0.5 s)	0.690460	0.646329	0.592194	0.688178	0.636588	0.693916
DeepLOB - snapshot only (1 s)	0.688503	0.651998	0.614595	0.695975	0.652758	0.700247
LOB-Transformer - snapshot only (0.5 s)	0.612476	0.580863	0.528183	0.645002	0.580776	0.616468
LOB-Transformer - snapshot only (1 s)	0.619193	0.579604	0.537200	0.761754	0.630067	0.621915
Deep LSTM - snapshot & physical measures (0.5 s)	0.693153	0.653711	0.601555	0.683185	0.639777	0.692689
Deep LSTM - snapshot & physical measures (1 s)	0.693166	0.660953	0.622677	0.707029	0.662177	0.704316
DeepLOB - snapshot & physical measures (0.5 s)	0.690947	0.648535	0.596498	0.677407	0.634383	0.695138
DeepLOB - snapshot & physical measures (1 s)	0.693882	0.661360	0.630580	0.674759	0.651922	0.709587
LOB-Transformer - snapshot & physical measures (0.5 s)	0.661615	0.596492	0.532882	0.839080	0.651812	0.676322
LOB-Transformer - snapshot & physical measures (1 s)	0.663810	0.635582	0.601237	0.666958	0.632395	0.679886

Several patterns emerge from Table 4. First, the baseline of physical momentum directly computed from Level-3 order flow without any optimisation has a performance only a few percentage lower than the best-performing deep learning architecture, despite requiring neither GPU training time nor hyperparameter search. It demonstrates that much of the short-term predictive signal in a limit order book is contained in mechanically interpretable quantities such as net momentum and energy, rather than in opaque non-linear transformations of raw depth snapshots.

Another observation is that deep learning on the physical measures alone surpasses the performance obtained from the raw depth snapshots in two cases. For example, the Deep LSTM fed only the four physics variables reaches 0.684 (0.5 s) and 0.685 (1 s) directional accuracy, beating its snapshot-only counterpart by roughly two percentage points. The LOB-Transformer benefits even more: physical-only training lifts its 1-second accuracy from 0.619 to 0.673, a gain of more than five percentage points. These results confirm that the physics features capture a large share of the short-horizon signal that is usually sought with much higher-dimensional inputs.

Second, augmenting snapshot tensors with four physical measures (energy of new order submissions, energy of cancellations, upward momentum and downward momentum) always improves the neural networks' out-of-sample scores. For the Deep LSTM model, sklearn accuracy rises from 0.636 to 0.654 (0.5 s horizon) and ROC AUC from 0.681 to 0.693, while directional accuracy increases by roughly 2 percentage points for every architecture tested. The

gain is largest for the LOB-Transformer, whose global self-attention evidently benefits from an explicit, physics-derived summary of recent order-flow stress. This synergy shows that the Physical Model is not merely a light-weight alternative to deep learning; it is a complementary feature generator that can be dropped into existing pipelines to boost performance with negligible additional latency.

Third, the table highlights a trade-off between raw headline accuracy and computational cost. All deep networks require tens of epochs of GPU training and non-trivial feature engineering (windowing, normalisation and label alignment). The Physical Model, by contrast, is trained on no data, while it merely requires the calculation of *Active Depth* in Section 4.2 which can be computed real-time with little computing power and time. The Physical Model is able to be executed in linear time with respect to the data stream, and produces interpretable alarms suitable for real-time surveillance systems. For practitioners concerned with computing power consumption, deployment complexity or model interpretability, the physical features offer a principled middle ground: they deliver much of the accuracy advantage of deep learning while preserving the transparency of the signal.

Finally, as the momentum baseline is computed in an event-driven manner and then resynchronised to an arbitrary snapshot clock - it is not tied to a particular sampling frequency. This is in stark contrast to supervised networks, whose weights are horizon-specific and must be re-trained if the prediction latency is required to be changed. Hence the Physical Model generalises across both temporal resolutions and market venues with minimal re-calibration.

These findings reinforce two findings of this study: (i) Purely statistical or deep learning baselines under-utilise microstructural information that is readily available in the message stream; the Physical Model captures that information explicitly. (ii) When the physical measures are supplied to the networks, their marginal benefit exceeds that of adding yet another layer or attention head, implying that future work should focus on richer feature engineering rather than ever-deeper architectures. Any comparative analysis that follows should therefore emphasise complementarity rather than a winner-takes-all narrative: the best practice is to couple lightweight, mechanistically grounded signals with data-hungry models when latency budgets permit.

Comparative Analysis

Efficiency and deployability: Deep architectures demand GPU hardware,

extensive hyper-parameter tuning and hours of training. By contrast, the Physical Model runs in real time on a standard CPU because its computations reduce to algebraic operations on incoming LOB updates. In latency-sensitive trading systems every millisecond counts; therefore, a model that incurs zero training time and minimal inference latency represents a substantial engineering advantage.

Data parsimony and robustness: Large-scale neural networks perform best when vast, stationary data sets are available. In volatile crypto markets, distributional shifts are frequent and can quickly erode learnt representations. The Physical Model sidesteps this fragility: its parameters are derived from first principles (trade-arrival rates, volume imbalance, queue dynamics) and update organically with the latest order-book state, maintaining relevance without re-training.

Interpretability and risk management: Regulatory and fiduciary constraints increasingly require transparent decision rules. The Physical Model's components (e.g. best-bid queue depletion, depth-weighted imbalance) map directly to observable microstructural phenomena, enabling traders and risk managers to trace a forecast back to concrete market mechanisms. In contrast, attention maps or LSTM cell activations rarely translate into actionable narratives.

Scalability across depths and assets: Neural networks typically truncate the order book to the top 10-20 levels for tractability. The Physical Model can ingest arbitrarily deep queues because its computational cost grows linearly with depth, not quadratically with parameter count. This flexibility is vital when quoting on venues where meaningful liquidity may reside dozens of ticks away. Moreover, because the model is data-agnostic, it transfers effortlessly from highly liquid BTC/USD to thinner alt-coin pairs without re-estimation.

Complementarity rather than competition: Finally, nothing precludes integrating approaches: the outputs of physical model can be appended as features to Transformer or LSTM inputs, supplying inductive bias that may accelerate convergence and improve stability. Our empirical results already hint at this synergy: augmenting snapshot data with physical measures raises the LSTM's one-second directional accuracy by 2.12% and boosts LOB-Transformer by 4.47%.

Order Book Depth and Price Dynamics: The study of Cont et al. (2014) discovered that while the first level has the strongest impact on price dynamics, deeper depths still contain information relevant for forecasting. The experimental findings from Sirignano and Cont (2019) and Zhang et al. (2019)

further underscored and proved the predictive significance of the deeper depths in the LOB on price dynamics. They defined and calculated sensitivity of a given depth in the LOB, which was 1.7% for the best bid/best ask, 0.4% for depth 2, 0.2% for both depth 3 and 4, and the total sensitivity of depth 5-10 was 0.4%. The truth is the depth beyond contain even more significant information, however, the restriction on computing power and model complexity of deep learning models make it difficult to train with the data for deeper depth. Contrarily, the Physical Model decides the depths relevant to the price dynamics by calculating the defined *Active Depth* in order to take full advantage. As a result, the Physical Model would be more efficient to capture the price dynamics compared to deep learning models.

In summary, while deep neural networks currently post the highest raw accuracies, the Physical Model offers a compelling blend of real-time operability, interpretability and cross-asset robustness. Its favourable cost-benefit profile makes it an indispensable component of any practical high-frequency forecasting toolkit.

9. Conclusion

This paper presents a novel framework for analyzing limit order book dynamics through a physics-inspired lens. By modelling individual orders as physical particles and conceptualising the order book as a physical system, we introduce two interpretable microstructural measures, kinetic energy and momentum, to characterise and forecast short-term price volatility and directional changes. Crucially, we define the concept of *Active Depth* to isolate the region of the order book where the most influential order activity occurs, allowing for efficient, high-resolution modelling without relying on time-based snapshots.

Our approach makes full use of Level-3 order book data, preserving temporal granularity and enabling a microscopic view of market dynamics. Through a case study of the LUNA cryptocurrency flash crash, we reveal contrasting behaviours between limit and market orders and offer empirical evidence that further supports the physical interpretation of market microstructure. The empirical analysis demonstrates that our physical model consistently outperforms established measures such as VPIN and Order Flow Imbalance (OFI), and rivals state-of-the-art deep learning architectures including Deep LSTM, DeepLOB, and Transformer-based models.

Beyond predictive performance, the Physical Model is computationally efficient, real-time deployable, scalable to deep levels of the LOB, and easily interpretable. Furthermore, we show that these physics-based measures enhance the accuracy of deep learning models when integrated as input features, underscoring their value not only as standalone tools but also as complementary signal generators.

This work highlights the potential of econophysics in the financial analysis of digital asset, and encourages a synthesis of physical intuition with modern machine learning. Future work could extend this framework to other asset classes such as equities, foreign exchanges and futures, or develop hybrid models that fuse physical and data-driven components for even more robust forecasting and interpretability. Together, these directions offer a compelling path toward a deeper, more structured understanding of financial market dynamics..

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